M13 Batch 9 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 768–917

Remote compass systems, glass cockpit and EFIS displays, central warning systems and the ground proximity warning system.

390 questions · Level 3: 130 · Level 2: 181 · Level 1: 79

● marks a question built on a fact your instructor highlighted or underlined in the manual (344 of 390). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Remote compass systems and slaving

PDF pages 771–787 · 72 questions

LEVEL 1 — RECALL

1. Where on an aeroplane are the flux valves located ●
   a. in the nose radome, in the forward cargo bay or in the wheel well
   b. at the wing tips, in the vertical stabiliser or in the tail cone
   c. under the cockpit floor, in the engine pylons or in the centre tank
2. Where is the flux valve of a helicopter usually installed ●
   a. under the cockpit floor, in the nose compartment or on the landing gear skid
   b. on the roof of the cabin, in the rotor head or on the upper deck of the engine
   c. on the bottom of the tail boom, the vertical stabilizers or in the vertical fin
3. A directional gyro is left running for a long flight leg. Which disadvantage shows up? ●
   a. a falsified reading in the region of the magnetic poles
   b. a falsified heading during accelerations and turns
   c. a drift of the indication as the time passes
4. Which disadvantage do magnet compasses and flux valves have ●
   a. a drift which grows with the running time of the unit
   b. a heading which lags behind by a fixed amount in the climb
   c. a falsified heading during accelerations and turns
5. What do the two acronyms in the name ADF RMI stand for ●
   a. Automatic Direction Finder and Radio Magnetic Indicator
   b. Airborne Direction Finder and Radio Magnetic Instrument
   c. Automatic Deviation Finder and Remote Magnetic Indicator
6. What do the arrows in the ADF RMI point to ●
   a. the stations that are tuned in
   b. the magnetic north found by the flux valve
   c. the course selected on the compass rose
7. Which three units does a remote compass with ADF/RMI have ●
   a. a flux valve, a directional gyro and an ADF/RMI
   b. a directional gyro, a compass coupler and a compass controller
   c. a flux valve, a compass coupler and an ADF/RMI

8. Where are the amplifiers that raise the voltage of weak signals installed ●
   a. in the integrated instrument system rack
   b. in the automatic direction finder indicator
   c. in the housing of the directional gyro

9. Which errors are adjusted by the screws on the flux valve compensation device ●
   a. the A and B errors
   b. the A error only
   c. the B and C errors

10. How is the A error of a remote compass system compensated ●
    a. by turning the remote compensator potentiometer
    b. by turning the whole flux valve
    c. by turning the compensation device screws

11. Which units make up a remote compass system with a compass coupler ●
    a. flux valve, DG, ADF RMI and compass coupler
    b. flux valve, DG, compass controller and remote compensator
    c. flux valve, DG, compass controller and compass coupler

12. With which control can any desired heading be set ●
    a. the DG – SLAVED switch
    b. the SET HDG knob
    c. the synchronizing knob

13. Fast synchronisation is unserviceable on a compass system. Where is the control logic for it held? ●
    a. in the compass controller
    b. in the directional gyro
    c. in the compass coupler

14. To which value does fast synchronisation raise the update rate of the magnetic heading ●
    a. 700 °/min.
    b. 70 °/min.
    c. 7 °/min.

LEVEL 2 — UNDERSTAND AND EXPLAIN

15. A flux valve is to go at a wing tip rather than near the cabin. Why is that place chosen? ●
    a. to keep them inside the pressurised area so they can be reached in flight
    b. to exclude the influence of magnetic interference fields as far as possible
    c. to bring them as close as possible to the heading indicators they supply

16. A helicopter flux valve is to be relocated and the shop asks what rules the choice of place. What governs it? ●
    a. it must be inside the cabin of the helicopter to keep the run short
    b. it must be close to the gearbox of the helicopter for a rigid mount
    c. it must be far away from all magnetic fields of the helicopter

17. Why is the heading of a flux valve falsified while the aircraft accelerates ●
    a. the valve loses part of its excitation while the aircraft is accelerating
    b. the valve swings so that it is also affected by the vertical magnetic field
    c. the valve output is damped so that it follows the change far too slowly

18. Which two units are combined to build a remote compass system that shows the correct heading ●

- **a.** a flux valve and a magnet compass
- **b.** a directional gyro and a magnet compass
- **c.** a flux valve and a directional gyro

19. From which unit does the indicator of a remote compass system receive the stable heading information ●

- **a.** from the directional gyro
- **b.** from the flux valve
- **c.** from the magnet compass

20. How is the difference between the magnetic field direction and the indicated heading used ●

- **a.** it drives the compass rose of the indicator directly
- **b.** it changes the excitation voltage of the flux valve
- **c.** it adjusts the DG through a torque motor

21. The slaving torque motor applies a torque to the gyro. What does that torque produce? ●

- **a.** a precession which changes the DG direction
- **b.** a friction brake which holds the DG direction
- **c.** a rigidity which locks the DG to magnetic north

22. A crew has one ADF RMI on the panel. Which two kinds of information does it give them? ●

- **a.** a radio navigation indication and the aircraft attitude
- **b.** the magnetic heading and the distance to the station
- **c.** a radio navigation indication and the magnetic heading

23. What does the slaving annunciator show the pilot ●

- **a.** the difference between the indicated heading and the flux valve signal
- **b.** the difference between the excitation voltage and its nominal value
- **c.** the difference between the selected course and the indicated heading

24. In the ADF/RMI system, where is the difference between the indicated heading and the flux valve signal detected

- **a.** inside the compass coupler
- **b.** inside the flux valve
- **c.** inside the indicator itself

25. A pilot reaches for the synchronizing knob on the ADF RMI. What can he change with it? ●

- **a.** the rate at which the torque motor runs
- **b.** the mode in which the system operates
- **c.** the indication of the compass rose

26. What is protected by the fuse fitted in the integrated instrument system rack ●

- **a.** the torque motor which is fitted inside the directional gyro
- **b.** the transformer which supplies the flux valve excitation
- **c.** the amplifier which raises the compass rose motor voltage

27. In which form does the detected field direction reach the control transformer in the indicator ●

- **a.** as a direct current signal
- **b.** as a single-phase signal
- **c.** as a three-phase signal

28. A slaving control transformer is fitted in an ADF/RMI. What is its rotor connected to? ●

a. the compass card
b. the synchronisation knob
c. the slaving torque motor

29. The north found by the flux valve matches the position of the compass rose. What appears at the slaving CT output? ●

a. a steady direct voltage
b. a signal of maximum size
c. no signal at all

30. In which condition of flight does the information from the flux valve contain no errors? ●

a. when flying at a constant height in any manoeuvre
b. when flying straight ahead without acceleration
c. when flying a steady turn at a constant bank angle

31. Which precision of heading indication is possible all the time with the remote compass system ●

a. +/− 0.5°
b. +/− 5°
c. +/− 2°

32. Where is the amplified error signal from the control transformer sent ●

a. to the slaving torque motor
b. to the compass rose motor
c. to the flux valve signal coils

33. After switch on the gyro is not aligned with magnetic north. How long can the automatic alignment take? ●

a. up to 90 minutes
b. up to 9 minutes
c. up to 90 seconds

34. The annunciator needle of the ADF/RMI sits at its central position. What does this mean? ●

a. the system is in the free azimuth mode
b. the slaving amplifier has failed
c. the slaving CD is zeroed out

35. What does the annunciator on the ADF/RMI display ●

a. the output voltage of the slaving amplifier
b. the output voltage of the slaving torque motor
c. the excitation voltage supplied to the flux valve

36. Between which two units can a remote compensator be switched ●

a. flux valve and directional gyro
b. flux valve and compass coupler
c. compass coupler and compass controller

37. A meter is put on the supply output of a remote compensator. Which kind of voltage is there?

a. a variable low frequency AC voltage
b. an extremely constant DC voltage
c. an extremely constant three-phase voltage

38. How is the compensation device carrying the B and C error adjustment screws fitted ●

- **a.** it is screwed into the compass coupler
- **b.** it is wired in series with the excitation coil
- **c.** it is screwed onto the flux valve

39. A compensation device is to be fitted to a flux valve. In which case is this done? ●

- **a.** only in the case of a failed remote compensator
- **b.** only in the case of persistent interference fields
- **c.** only in the case of a drifting directional gyro

40. A newer type of compass system is fitted. Where is the fuse for the flux valve excitation transformer additionally installed?

- **a.** in the flux valve housing
- **b.** in the compass coupler
- **c.** in the compass controller

41. The synchronising annunciator is deflected. Along which path is the difference being reported? ●

- **a.** flux valve – coupler – DG
- **b.** DG – coupler – autoflight system
- **c.** flux valve – coupler – HSI

42. Which two signals are coupled together in the compass coupler in the slaved mode ●

- **a.** the DG platform heading and the autoflight heading
- **b.** the flux valve signal and the DG platform heading
- **c.** the flux valve signal and the selected VOR course

43. Flux valve and gyro headings disagree in a coupler system. Which device balances them out? ●

- **a.** a control transformer
- **b.** a differential synchro
- **c.** a slaving torque motor

44. A compass coupler system is compared with a system using an ADF/RMI. What does the comparison show? ●

- **a.** the same indication behaviour and the same precision
- **b.** a worse indication behaviour but a higher precision
- **c.** a better indication behaviour but a lower precision

45. The system is in its run-up phase. What happens to the update rate of the differential synchro? ●

- **a.** it is held at 2°/min.
- **b.** it is increased to 700°/min.
- **c.** it is reduced to 0.2°/min.

46. Which source provides the directional information in the DG mode ●

- **a.** only the short-term stable gyro platform
- **b.** only the long-term stable flux valve
- **c.** the gyro platform and the flux valve together

47. What is the state of the synchronising annunciator in the DG mode ●

- **a.** it stands at its centre
- **b.** it shows the drift rate
- **c.** it is switched off

48. How does the fast synchronisation control logic raise the update rate ●

- **a.** by bypassing the differential synchro
- **b.** by switching in a second torque motor
- **c.** by switching the amplification factor

49. After which length of switch-off pause does switching on in SLAVED mode start the fast synchronisation

- **a.** about 10 seconds
- **b.** about 2 seconds
- **c.** about 2 minutes

LEVEL 3 — APPLY AND REASON

50. What is gained by holding the torquing rate of the directional gyro to its printed maximum ●

- **a.** drift is still compensated but accelerations cannot move the indication
- **b.** alignment is faster but the gyro cannot be moved by its own drift
- **c.** turns are still compensated but drift cannot move the indication

51. An aircraft is in a long accelerating climb and the flux valve signal is briefly false. Why does the indicated heading stay good? ●

- **a.** the torque motor reverses and drives the gyro back to its old heading
- **b.** the gyro can be torqued only slowly, so a brief error cannot move it
- **c.** the flux valve signal is cut off automatically whenever speed changes

52. In the cruise the slaving annunciator keeps fluctuating around zero. What does this tell the pilot?

- **a.** the gyro is still running up and the heading is not yet usable at all
- **b.** the gyro has lost its stable direction and the valve is being ignored
- **c.** the gyro has the correct stable direction and the valve is correcting it

53. The slaving annunciator shows a long term deflection to one side. What should be concluded?

- **a.** the flux valve is seeing an acceleration error
- **b.** there is an error in the system
- **c.** the gyro drift is being corrected normally

54. Why can the compass rose of the ADF RMI show any heading immediately on power up ●

- **a.** the direction taken up by the gyro after run-up may be any direction
- **b.** the slaving annunciator is held at zero during the whole run-up
- **c.** the flux valve needs several minutes before it delivers any signal

55. The system has just been switched on and the heading shown is wrong. Why does it take time to become correct? ●

- **a.** the compass rose motor is deliberately damped during the run-up
- **b.** the slaving torque motor may move the gyro by only 2° per minute
- **c.** the flux valve must first warm up before it delivers its excitation

56. A pilot uses the synchronizing knob shortly after start up. What is the effect? ●

- **a.** the time until the heading indication is correct becomes much shorter
- **b.** the torquing rate of the gyro is raised to 700° per minute
- **c.** the slaving annunciator is switched off for the rest of the flight

57. Why is the flux valve signal coupled with the gyro heading instead of being sent straight to the indicator ●

- **a.** it can fluctuate wildly for a short time under the influence of acceleration
- **b.** it is far too weak to drive the compass rose motor of the indicator
- **c.** it is a three-phase signal which the compass card cannot accept at all

58. Which kind of stability is supplied by each unit of a remote compass with ADF/RMI ●

- **a.** long-term stability from the gyro, short-term from the flux valve
- **b.** long-term stability from the flux valve, short-term from the gyro
- **c.** long-term stability from the coupler, short-term from the gyro

59. An aircraft accelerates on take off and then cruises for hours. Which unit takes out which error? ●

- **a.** the coupler takes out the acceleration errors and the valve the drift
- **b.** the valve takes out the acceleration errors and the gyro the drift
- **c.** the gyro takes out the acceleration errors and the valve the drift

60. An engineer does not want to wait for the automatic alignment. How is the stator of the control transformer adjusted? ●

- **a.** by holding the mode switch in the SLAVED position
- **b.** by depressing and rotating the synchronisation knob
- **c.** by depressing and rotating the course selector knob

61. Which errors are offset by the two precision potentiometers of the remote compensator

- **a.** the A and B errors
- **b.** the A error alone
- **c.** the B and C errors

62. A flux valve is indexed, that is prefixed. What does this mean when the A error is to be compensated?

- **a.** it must be twisted until the B error is also at zero
- **b.** it must not be twisted in its mounting
- **c.** it must be twisted only with the compensator switched off

63. What makes it unnecessary to update the gyro by torquing in a system with a compass coupler ●

- **a.** the differences are balanced out by a differential synchro
- **b.** the gyro used in such a system has no drift to be corrected
- **c.** the flux valve signal is passed to the indicator on its own

64. What becomes of the two signals in the coupler when the DG mode is selected ●

- **a.** the heading signal is switched off, the valve signal still passes through
- **b.** the heading signal still passes through, the valve signal is amplified more
- **c.** the heading signal still passes through, the valve signal is switched off

65. The flux valve fails during flight. Which mode of operation has to be used? ●

- **a.** the fast synchronisation mode
- **b.** the DG mode
- **c.** the slaved mode

66. The system has been in DG mode for a long time and the heading has slowly wandered. What must the crew do? ●

- **a.** switch briefly to the slaved mode and straight back again
- **b.** wait for the synchronising annunciator to null it out
- **c.** correct the indication with the SET HDG knob

67. A remote compass system is found in which the gyro is never torqued. Which arrangement must it have? ●

- **a.** a coupler in which a differential synchro balances the signals
- **b.** an indicator in which a control transformer balances the signals
- **c.** a coupler in which a torque motor balances the signals

68. Fast synchronisation has been running for well over its minimum time. What else must happen before it is complete?

- **a.** the synchronisation error must have come down to 2° or less
- **b.** the synchronising annunciator must have been off for 2 minutes
- **c.** the synchronisation error must have come down to 2° per minute

69. Which state of the fast synchronisation is required for the superflag signal to appear

- **a.** fast synchronisation must have been run at least once
- **b.** fast synchronisation must not be in operation
- **c.** fast synchronisation must be in operation at the time

70. Which two conditions concerning the gyro and the servo circuit must hold for the superflag signal to appear

- **a.** the DG or INS is valid and the servo circuit is nulled out
- **b.** the DG is in free azimuth and the servo circuit is complete
- **c.** the DG or INS is valid and the servo circuit is complete

71. The maintenance monitor responded only briefly during a flight. What does that mean for the compass system?

- **a.** its functioning is not affected
- **b.** its heading output must be treated as invalid
- **c.** it must be put into the DG mode for the rest of the flight

72. A compass system is switched off for half a minute and then switched on again in SLAVED. Does fast synchronisation run? ●

- **a.** no, the switch-off pause was too short for it
- **b.** yes, every switch-on in SLAVED will start it
- **c.** no, it runs only when the mode is left in DG

Answers — Remote compass systems and slaving

1. **b** Those three positions are the ones used for the flux valve on an aeroplane. (L1, PDF p. 771)

2. **c** Those are the usual installation places for the flux valve of a helicopter. (L1, PDF p. 773)

3. **c** Drift is the disadvantage of the gyro based direction instruments. (L1, PDF p. 775)

4. **c** Accelerations and turns are what falsify the magnetic sensing devices. (L1, PDF p. 775)

5. **a** ADF is Automatic Direction Finder and RMI is Radio Magnetic Indicator. (L1, PDF p. 777)

6. **a** The ADF arrows point at whichever broadcast stations have been tuned. (L1, PDF p. 777)

7. **a** Those three are the units listed for the remote compass with ADF/RMI. (L1, PDF p. 779)

8. **a** Weak signals such as the compass rose motor voltage are amplified in the IIS rack. (L1, PDF p. 779)

9. **c** The screws on the compensation device take out the B and C errors. (L1, PDF p. 781)

10. **b** The A error is taken out by rotating the valve itself, not by a screw or a potentiometer. (L1, PDF p. 781)

11. **c** Those four are the units listed for the coupler based system. (L1, PDF p. 785)

12. **b** The SET HDG knob is the one for setting a heading of choice. (L1, PDF p. 785)

13. **c** The fast synchronisation logic sits inside the coupler. (L1, PDF p. 787)

14. **a** Fast synchronisation runs at seven hundred degrees a minute. (L1, PDF p. 787)

15. **b** The locations are chosen to keep magnetic interference fields away from the valve. (L2, PDF p. 771)

16. **c** Distance from the helicopter's own magnetic fields decides where the flux valve goes. (L2, PDF p. 773)

17. **b** The swing lets the vertical component of the earth's field reach the valve. (L2, PDF p. 775)

18. **c** The flux valve and the directional gyro together make the remote compass system. (L2, PDF p. 775)

19. **a** Stable heading comes to the indicator from the directional gyro. (L2, PDF p. 775)
20. **c** The difference signal is fed to a torque motor which moves the gyro. (L2, PDF p. 775)
21. **a** Torque on a gyro acts as precession and so moves the gyro's direction. (L2, PDF p. 775)
22. **c** The one indicator carries both the radio navigation picture and the magnetic heading. (L2, PDF p. 777)
23. **a** The annunciator displays how far the indicated heading is from what the flux valve says. (L2, PDF p. 777)
24. **c** In this architecture the detection happens in the indicator itself. (L2, PDF p. 777)
25. **c** The knob acts on what the compass rose shows. (L2, PDF p. 777)
26. **b** That fuse is there for the excitation transformer of the flux valve. (L2, PDF p. 779)
27. **c** The flux valve passes its direction information on as a three-phase signal. (L2, PDF p. 779)
28. **a** The rotor of the slaving CT is driven directly by the compass card. (L2, PDF p. 779)
29. **c** Agreement between the two gives a null at the control transformer. (L2, PDF p. 779)
30. **b** Unaccelerated straight flight is the condition in which the valve is error free. (L2, PDF p. 779)
31. **c** Two degrees either way is the precision of the remote compass system. (L2, PDF p. 779)
32. **a** The amplified error signal drives the slaving torque motor on the gyro. (L2, PDF p. 779)
33. **a** Working off a large error at the slow torquing rate can take as long as ninety minutes. (L2, PDF p. 779)
34. **c** Centre position of the moving coil annunciator means the slaving CD is nulled. (L2, PDF p. 779)
35. **a** The moving coil annunciator reads the slaving amplifier output. (L2, PDF p. 779)
36. **b** The remote compensator sits in the line between the flux valve and the coupler. (L2, PDF p. 781)
37. **b** The compensating supply is a very constant direct voltage. (L2, PDF p. 781)
38. **c** The compensation device is mounted directly on the flux valve. (L2, PDF p. 781)
39. **b** It is reserved for interference fields that will not go away. (L2, PDF p. 781)
40. **b** On newer couplers the protecting fuse is also carried in the coupler. (L2, PDF p. 785)
41. **a** It watches the chain from the valve through the coupler to the gyro heading. (L2, PDF p. 785)
42. **b** Slaved mode brings the valve signal and the gyro heading together in the coupler. (L2, PDF p. 785)
43. **b** The differential synchro in the coupler takes out the difference. (L2, PDF p. 785)
44. **a** There is no difference between the two systems in indication behaviour or in precision. (L2, PDF p. 785)
45. **b** Run-up uses the much higher rate so the system lines up quickly. (L2, PDF p. 785)
46. **a** In DG mode the gyro alone supplies the direction. (L2, PDF p. 785)
47. **c** There is nothing for it to compare in DG mode, so it is switched off. (L2, PDF p. 785)
48. **c** A change of amplification factor is what gives the higher rate. (L2, PDF p. 787)
49. **c** A pause of roughly two minutes before a SLAVED start triggers the logic. (L2, PDF p. 787)
50. **a** The rate is fast enough for drift but too slow for the short acceleration errors to get through. (L3, PDF p. 775)
51. **b** The slow torquing rate means a short lived false signal has no time to shift the indication. (L3, PDF p. 775)
52. **c** Fluctuation about zero is the sign of a healthy system doing its normal correcting. (L3, PDF p. 777)
53. **b** A deflection that lasts, rather than one that fluctuates, means a system error. (L3, PDF p. 777)
54. **a** The rose follows the gyro, and the gyro runs up pointing anywhere. (L3, PDF p. 777)
55. **b** A large initial error can only be worked off at the slow torquing rate. (L3, PDF p. 777)
56. **a** Minimising the difference by hand saves the slow automatic torquing. (L3, PDF p. 777)
57. **a** Acceleration makes the raw valve signal jump about, so the gyro steadies it. (L3, PDF p. 779)
58. **b** The valve holds the long term truth, the gyro the short term steadiness. (L3, PDF p. 779)
59. **c** The gyro covers the short acceleration errors, the flux valve the slow drift. (L3, PDF p. 779)
60. **b** The stator is set by hand with the synchronisation knob. (L3, PDF p. 779)
61. **c** The N − S and E − W settings together deal with the B and C errors. (L3, PDF p. 781)
62. **b** An indexed valve is not allowed to be turned, so that way of taking out the A error is barred. (L3, PDF p. 781)
63. **a** Because the coupler itself removes the difference, the gyro need not be torqued. (L3, PDF p. 785)

64. **c** Only the flux valve signal is cut; the heading still runs through the coupler. (L3, PDF p. 785)

65. **b** With no usable valve signal the system has to run on the gyro alone, which is DG mode. (L3, PDF p. 785)

66. **c** In DG mode nothing corrects the drift automatically, so it is set out by hand. (L3, PDF p. 785)

67. **a** Only the coupler arrangement with a differential synchro removes the need to torque the gyro. (L3, PDF p. 785)

68. **a** Both the time and an error within two degrees are needed before it finishes. (L3, PDF p. 787)

69. **b** One of the listed conditions for the superflag is that fast synchronisation is not running. (L3, PDF p. 787)

70. **c** Validity of the gyro or INS and a complete servo circuit are two of the listed conditions. (L3, PDF p. 787)

71. **a** A temporary response has no effect on how the system works. (L3, PDF p. 787)

72. **a** A pause of about two minutes is needed, so half a minute does not trigger the logic. (L3, PDF p. 787)

Heading indicators and system interfaces

PDF pages 789–795 · 21 questions

LEVEL 1 — RECALL

73. What shows the pilot that the signal from his compass system is not valid ●

a. a heading warning flag
b. the synchronising annunciator
c. the compass superflag

74. What does the abbreviation HSI stand for ●

a. Horizontal Situation Indicator
b. Heading and Situation Instrument
c. Horizontal Stabiliser Indicator

75. A pilot glances at the HSI for his heading. Which indication on it is the main one? ●

a. the distance readout
b. the course bar
c. the compass rose

LEVEL 2 — UNDERSTAND AND EXPLAIN

76. A pilot reads the heading on the ND in front of him. Which compass system supplies it? ●

a. the opposite compass system
b. their own compass system
c. whichever system the transfer switch selects

77. Which compass system do the RMIs show ●

a. the system feeding the autoflight computer
b. the opposite compass system
c. their own compass system

78. To which autoflight system is the heading of compass system 1 provided ●

a. to autoflight system number 1
b. to autoflight system number 2
c. to both autoflight systems together

79. To what besides the other heading sensor can the heading transfer switch connect a user ●

a. to a third unit if one is available
b. to the flux valve signal taken on its own
c. to the autoflight system of the other side

80. A heading leaves the compass coupler on its way to the HSIs. Which heading is it?

a. the uncorrected flux valve heading
b. the gyro supported magnetic heading
c. the free azimuth platform heading

81. To which compass system is the flight recorder normally switched ●

a. to compass system 1
b. to both systems in turn
c. to compass system 2

82. How are the ADF RMIs normally switched to the two compass systems? ●

- **a.** straight, each to its own system
- **b.** crosswise, each to the opposite system
- **c.** both to the same system

83. From which viewpoint does the HSI present the situation of the aircraft ●

- **a.** as seen from behind
- **b.** as seen from above
- **c.** as seen from the side

84. To which kind of ground station is the course selected on an HSI related ●

- **a.** an ADF broadcast station
- **b.** a VOR ground radio station
- **c.** a DME ground station

85. The white bar on an HSI has moved away from the centre. What is it telling the crew? ●

- **a.** there is a deviation from the selected heading
- **b.** there is a deviation from the selected course
- **c.** there is a deviation in the distance to the station

86. What can the HSI show with respect to a DME ground station ●

- **a.** the bearing to it
- **b.** the distance to it
- **c.** the deviation from it

87. Which kind of system do all the further indications on an HSI belong to ●

- **a.** autoflight systems
- **b.** navigation systems
- **c.** attitude reference systems

LEVEL 3 — APPLY AND REASON

88. A pilot compares the heading on the ND with the heading on the RMI in front of him. Which sources is he comparing? ●

- **a.** the opposite compass system against the flight recorder
- **b.** his own compass system against the autoflight system
- **c.** his own compass system against the opposite one

89. A heading warning flag has appeared on one side, so what can the pilot do about it? ●

- **a.** use the compass controller to take the free azimuth heading instead
- **b.** use the synchronizing knob to re-set the compass rose by hand
- **c.** use the heading transfer switch to take the other heading sensor

90. An HSI loses its heading while the RMI beside it still shows one. Which compass system has gone? ●

- **a.** the one selected with the heading transfer switch
- **b.** the one belonging to that side of the cockpit
- **c.** the one belonging to the opposite side of the cockpit

91. One of the two compass systems fails in flight. What happens to the consumers that were fed by it?

- **a.** they can all be switched to the system that is still working
- **b.** they are fed directly from the flux valve of the failed system
- **c.** they hold the last valid heading until the system is reset

92. What is achieved by exciting the synchros from various voltage sources

- **a.** a receiver is always on a transmitting synchro fed by its own bus
- **b.** a receiver is always fed from two independent buses at the same time
- **c.** a receiver is always supplied even when the coupler itself has failed

93. An HSI shows several indications driven from different sources. Which one of them is driven by the heading system? ●

- **a.** the compass rose
- **b.** the course bar
- **c.** the distance

Answers — Heading indicators and system interfaces

73. **a** Loss of validity is shown by the heading warning flag. (L1, PDF p. 789)

74. **a** HSI is the Horizontal Situation Indicator. (L1, PDF p. 795)

75. **c** The compass rose is the HSI's main indication. (L1, PDF p. 795)

76. **b** The HSI or ND is fed by the compass system of its own side. (L2, PDF p. 789)

77. **b** The RMI deliberately carries the heading of the other side's system. (L2, PDF p. 789)

78. **a** The autoflight systems are fed straight across, one for one. (L2, PDF p. 789)

79. **a** Where a third unit exists the transfer switch can reach it too. (L2, PDF p. 789)

80. **b** What leaves the coupler is the magnetic heading after the gyro has steadied it. (L2, PDF p. 791)

81. **c** The flight recorder normally sits on compass system 2. (L2, PDF p. 791)

82. **b** The ADF RMIs are fed across, each from the other side's system. (L2, PDF p. 791)

83. **b** The HSI gives a plan view, the aircraft seen from above. (L2, PDF p. 795)

84. **b** The selectable course on the HSI belongs to a VOR station. (L2, PDF p. 795)

85. **b** The bar is the course deviation indication of the HSI. (L2, PDF p. 795)

86. **b** From a DME station the HSI takes the distance. (L2, PDF p. 795)

87. **b** Every extra indication on the HSI is a navigation system indication. (L2, PDF p. 795)

88. **c** Because the two displays are fed from opposite systems, the comparison is a cross check of both. (L3, PDF p. 789)

89. **c** The transfer switch is the way of getting a valid heading back. (L3, PDF p. 789)

90. **b** The HSI is fed by its own system and the RMI by the other one, so the failure is on its own side. (L3, PDF p. 789)

91. **a** Every affected consumer can be moved over to the surviving system. (L3, PDF p. 791)

92. **a** Spreading the excitation keeps each receiver and its synchro on a common bus. (L3, PDF p. 791)

93. **a** The rose is the heading indication; the course bar and the distance are navigation system indications. (L3, PDF p. 795)

Glass cockpit displays

PDF pages 797–807 · 41 questions

LEVEL 1 — RECALL

94. What is meant by a glass cockpit ●

a. the instrument glasses are replaced by a scratch free type
b. computers and displays are used instead of instruments
c. the instruments are fitted behind one large glass panel

95. What is the system used for flight monitoring in a glass cockpit called in many aircraft ●

a. EFIS
b. CDS
c. EIS

96. Which three items are the components of a typical electronic instrument system ●

a. Display units, control panels and pressure sensors
b. Display computers, display units and control panels
c. Display computers, control panels and warning horns

97. A system is fitted with as few display computers as it is allowed to have, so how many is that? ●

a. Three
b. Four
c. Two

98. How many indication display screens does the display system have in the cockpit ●

a. Eight
b. Six
c. Four

99. What do the letters ECAM stand for ●

a. Electronic centralized aircraft monitor
b. Electronic central alert monitor
c. Engine centralized aircraft monitor

100. How many display management computers are installed and what do they do with the data they receive ●

a. Three computers, forwarding the data to the display units
b. Two computers, forwarding the data to the display units
c. Three computers, forwarding the data to the warning computers

101. An engineer counts the flight warning computers of the system, so how many does he find? ●

a. One
b. Three
c. Two

LEVEL 2 — UNDERSTAND AND EXPLAIN

102. A crew is flying an aircraft with a glass cockpit. How is their flight monitoring done? ●

a. by one display for each pilot
b. by two displays for each pilot
c. by two displays shared by both pilots

103. Where are the many engine parameters shown in a glass cockpit ●

a. on the related engine control panel
b. on two displays for each pilot
c. on two central displays

104. Where are the alerts of the central warning panel presented in a glass cockpit ●

a. at the related control panel
b. in a display
c. on a printed caption board

105. The engine data and the auto flight system data both reach a display computer, so in which form do they arrive?

a. One is digital and one is analogue
b. Both are analogue
c. Both are digital

106. Why does a large aircraft carry a third display computer ●

a. To drive the control panels
b. For redundancy
c. To convert analogue inputs

107. How does the electronic flight instrument system stand in relation to the electronic instrument system ●

a. It is a part of it and shows navigation data
b. It is a separate system and shows engine data
c. It replaces it and shows alert messages

108. How many control panels serve the centre displays and what is selected on them ●

a. One panel, used to select the computer to be monitored
b. One panel, used to select the system to be monitored
c. Two panels, used to select the system to be monitored

109. Which further information is presented on the electronic instruments besides the navigation data ●

a. The limits of the engine parameters
b. The condition of the air conditioning system
c. The activities of the auto flight system

110. How are the six indication display screens divided between the electronic flight instruments and the central monitoring displays ●

a. Two flight instrument displays and four monitoring displays
b. Three flight instrument displays and three monitoring displays
c. Four flight instrument displays and two monitoring displays

111. Which two displays make up the centralized monitoring pair on an Airbus aircraft ●

a. The primary flight display and the system display
b. The engine and warning display and the system display
c. The engine and warning display and the navigation display

112. Which of the two EICAS displays on a Boeing aircraft carries the engine and warning indications ●

a. The upper EICAS display
b. The lower EICAS display
c. The auxiliary EICAS display

113. Which kind of information does the primary flight display supply ●

a. Flight planning information and parameters
b. Engine monitoring information and parameters
c. Flight monitoring information and parameters

114. Which of these parameters does the primary flight display supply? ●

a. Total air temperature
b. Gross weight
c. Radio altitude

115. Which annunciation does the primary flight display supply ●

a. The engine mode annunciation
b. The flight phase annunciation
c. The flight mode annunciation

116. With respect to which plane does the navigation display present the situation of the aircraft ●

a. The horizontal plane with respect to the flight path
b. The vertical plane with respect to the earth's surface
c. The horizontal plane with respect to the earth's surface

117. A flight warning computer works on the data it receives, so which three kinds of output does it generate? ●

a. Alert messages, aural alerts and synthetic voice
b. Alert messages, synthetic voice and engine parameters
c. Alert messages, aural alerts and system pages

118. An announcement in spoken words is heard in the cockpit, so which of the three warning outputs is this? ●

a. Synthetic voice
b. Aural alert
c. Alert message

119. Which computer decides what is to appear on the centralized monitoring displays ●

a. The flight warning computer
b. The system data acquisition concentrator
c. The display management computer

120. To which two sets of computers does a system data acquisition concentrator forward its data ●

a. The flight warning computers and the other concentrator
b. The display management computers and the control panels
c. The flight warning computers and the display management computers

121. Which computer puts the engine parameters and the system page parameters onto the displays? ●

a. The flight warning computer
b. The system data acquisition concentrator
c. The display management computer

122. Which three kinds of warning are generated by the modularized avionics warning electronics assembly ●

a. Optical, tactile and synoptic warnings
b. Optical, aural and visual warnings
c. Optical, aural and tactile warnings

123. Through which item of equipment are the warnings of the avionics warning electronics assembly generated ●

a. Through the control panel
b. Through the display select panel
c. Through the interface unit

124. Which three items are controlled by the avionics warning electronics assembly itself? ●

a. The master warning light, the aural warn speaker and the stick shaker
b. The master caution light, the aural warn speaker and the stick shaker
c. The master warning light, the aural warn speaker and the display units

LEVEL 3 — APPLY AND REASON

125. An aircraft without a glass cockpit is being compared with one that has it. What does flight monitoring use on the older aircraft? ●

a. two central displays and the indicators of a central warning panel
b. two displays for each pilot and some other navigation indicators
c. the basic T instruments and some other navigation indicators

126. On which display is the crew told that a value somewhere in the aircraft has gone out of limit

a. On the navigation display of each pilot
b. On one of the two engine and system data displays
c. On the primary flight display of each pilot

127. An aircraft is fitted with only the minimum number of display computers, so what does it lack compared with a large aircraft? ●

a. The computer that is there for redundancy
b. The computer that handles the analogue inputs
c. The computer that drives the centre displays

128. Which part inside a second generation display unit makes a picture out of the data it is sent

a. Its integrated symbol generator
b. Its integrated control panel
c. Its integrated display computer

129. Which Boeing display does the same job as the Airbus system display ●

a. The engine and warning display
b. The auxiliary EICAS display
c. The upper EICAS display

130. Which path is taken by the signal that produces an amber caution ●

a. From the sensors and computers through the data concentrator to the warning computer
b. From the sensors and computers direct to the warning computer
c. From the sensors and computers through the display computer to the warning computer

131. Both system data acquisition concentrators are lost, so which colour of warning can still be generated? ●

a. Neither colour
b. The red warnings
c. The amber cautions

132. The engine and warning screen stays blank but the aural alerts are still heard, so which computer is suspected? ●

- **a.** The display management computer
- **b.** The flight warning computer
- **c.** The data acquisition concentrator

133. The stick shaker does not operate although the conditions for it are met, so which unit is first suspected? ●

- **a.** The EICAS display select panel
- **b.** The avionics warning electronics assembly
- **c.** The EFIS and EICAS interface unit

134. The master warning light comes on but no message reaches the screens, so which item is suspected? ●

- **a.** The EICAS display select panel
- **b.** The avionics warning electronics assembly
- **c.** The EFIS and EICAS interface unit

Answers — Glass cockpit displays

94. **b** Glass cockpit means instruments have given way to computers and displays. (L1, PDF p. 797)

95. **a** The flight monitoring displays are commonly called EFIS. (L1, PDF p. 797)

96. **b** The three listed components are the display computers, the display units and the control panels. (L1, PDF p. 799)

97. **c** There are at least two display computers, and large aircraft have a third for redundancy. (L1, PDF p. 799)

98. **b** The cockpit carries six indication display screens. (L1, PDF p. 801)

99. **a** ECAM is the electronic centralized aircraft monitor. (L1, PDF p. 801)

100. **a** Three identical display management computers take in the data and pass it to the display units. (L1, PDF p. 805)

101. **c** Two identical flight warning computers are fitted. (L1, PDF p. 805)

102. **b** Each pilot has two displays for flight monitoring. (L2, PDF p. 797)

103. **c** Engine parameters go on the two central displays. (L2, PDF p. 797)

104. **b** Those alerts move from the panel onto a display. (L2, PDF p. 797)

105. **c** Navigation, auto flight and engine inputs are all listed as digital. (L2, PDF p. 799)

106. **b** The third computer is fitted on large aircraft for redundancy. (L2, PDF p. 799)

107. **a** EFIS is part of the EIS and its job is to show navigation data. (L2, PDF p. 799)

108. **b** A single control panel serves the centre displays and selects the system to be monitored. (L2, PDF p. 799)

109. **c** The electronic flight instruments also show what the auto flight system is doing. (L2, PDF p. 799)

110. **c** Four of the six are flight instrument displays and two are central monitoring displays. (L2, PDF p. 801)

111. **b** The pair is the engine and warning display together with the system display. (L2, PDF p. 801)

112. **a** The upper EICAS is the engine and warning display and the auxiliary one is the system display. (L2, PDF p. 801)

113. **c** The primary flight display is the flight monitoring display. (L2, PDF p. 801)

114. **c** Radio altitude is one of the parameters listed for the primary flight display; total air temperature and gross weight are carried on the engine and system displays instead. (L2, PDF p. 801)

115. **c** The flight mode annunciation is one of the listed primary flight display parameters. (L2, PDF p. 801)

116. **c** The navigation display works in the horizontal plane referred to the earth's surface. (L2, PDF p. 801)

117. **a** The three outputs are the optical alert messages, the acoustic aural alerts and the voice announcements. (L2, PDF p. 805)

118. **a** The voice announcement is the synthetic voice output, while the acoustic one is the aural alert. (L2, PDF p. 805)

119. **a** The flight warning computer determines the content of the ECAM displays. (L2, PDF p. 805)

120. **c** The concentrator feeds both the flight warning computers and the display management computers. (L2, PDF p. 805)

121. c Overlaying the engine and system page parameters is the display management computer's job. (L2, PDF p. 805)

122. c It produces optical, aural and tactile warnings. (L2, PDF p. 807)

123. c The warnings are generated by way of the EFIS/EICAS interface unit. (L2, PDF p. 807)

124. a Those three are the items it drives directly. (L2, PDF p. 807)

125. c Without displays the job is done by the basic T instruments plus other navigation indicators. (L3, PDF p. 797)

126. b One of the two engine and system displays additionally carries the alerts for out of limit conditions. (L3, PDF p. 799)

127. a The minimum fit is two computers, so the third one carried for redundancy on large aircraft is missing. (L3, PDF p. 799)

128. a These units are sent data rather than video, so the symbol generator built into them draws the picture. (L3, PDF p. 799)

129. b The auxiliary EICAS does the job that the system display does on the other manufacturer's fit. (L3, PDF p. 801)

130. a An amber caution reaches the flight warning computer by way of the system data acquisition concentrator. (L3, PDF p. 805)

131. b Red warnings reach the warning computer without passing through the concentrator, so they survive its loss. (L3, PDF p. 805)

132. a The warnings are still being generated, so the fault lies in the computer that feeds the display unit. (L3, PDF p. 805)

133. b The stick shaker is one of the items the warning electronics assembly controls directly. (L3, PDF p. 807)

134. c The light is driven directly and works, so the path through the interface unit to the screens is at fault. (L3, PDF p. 807)

Display management and failures

PDF pages 811–823 · 24 questions

LEVEL 1 — RECALL

135. What appears on the display in place of a parameter whose input signal is not available ●

a. A flag
b. An amber box
c. A dash

LEVEL 2 — UNDERSTAND AND EXPLAIN

136. What does a cathode ray tube display unit do first when it detects an internal overheat ●

a. It switches the display off
b. It switches the raster colours off
c. It switches the strokes off

137. What is left on the screen after the first step of the overheat reaction ●

a. Only strokes, with the warnings removed
b. Only strokes, with all information still given
c. Only the raster, with all information still given

138. Why does switching the raster colours off help a display unit that is running hot? ●

a. Less light is needed from the backlight
b. Less current is drawn by the cooling fan
c. Less heat is produced then

139. What does the display unit do as the second step of its overheat reaction ●

a. It switches the display to the cross side computer
b. It switches the raster colours off
c. It switches the display off

140. Which two conditions can cause an overheat of a display unit? ●

a. An internal failure or a problem in the cooling system
b. An internal failure or a problem in the power supply
c. A high cabin temperature or a problem in the cooling system

141. What happens to the white in the picture when one electron beam of a cathode ray tube fails ●

a. There is no white in the picture
b. The white is replaced by amber in the picture
c. The white turns to grey in the picture

142. What becomes of the information that is normally presented in the colour of a failed electron beam ●

a. It will not be shown
b. It is shown in white instead
c. It is shown as a flag instead

143. What does the display do when the red electron beam fails ●

a. It turns monochrome
b. It loses only the white
c. It switches itself off

144. Which electron beam has failed when the crew sees the whole display in one colour ●

a. The red beam
b. The green beam
c. The blue beam

145. Which part of the display is affected when one input signal to a display unit is lost ●

a. Only the warning labels of the display
b. Only the indication of the related parameter
c. The whole of the display unit

LEVEL 3 — APPLY AND REASON

146. On which panels does a helicopter crew find the engine alerts and the navigation picture

a. The alert on the centre panel and the picture on the outboard panels
b. The alert on the outboard panels and the picture on the centre panel
c. Both of them on the centre instrument panel

147. Where does the information of a primary flight display appear when that display goes blank in flight

a. On the navigation display
b. On the opposite primary flight display
c. On the centre engine display

148. A pilot's primary flight display goes dark, so what must he do before the information comes back?

a. Nothing, because the transfer happens on its own
b. Select the cross side computer on his control panel
c. Nothing, because the failed screen restarts on its own

149. A display unit that switched itself off because of overheat has cooled down again, so what happens next? ●

a. The picture comes back without any crew action
b. The picture comes back without the raster colours
c. The picture stays off until the crew resets it

150. Which failure fits a screen that has lost its colours but still shows every item of information ●

a. A total loss of the display unit
b. The second step of the overheat reaction
c. The first step of the overheat reaction

151. What comes next when a screen has already dropped its raster colours and the cause has not been cleared ●

a. The screen switches to the cross side computer
b. The screen switches itself off
c. The screen drops its strokes as well

152. A pilot reports that the green symbols have gone and nothing on the screen looks white any more, so what has happened? ●

a. The raster colours have been switched off
b. An input signal to the display has been lost
c. One electron beam of the tube has failed

153. Why is the display turned monochrome rather than left with its remaining colours when the red beam fails ●

a. Because the warning labels are red and must not be allowed to disappear
b. Because the warning labels are red and must not be allowed to flicker
c. Because the remaining colours would produce too much heat

154. Why is a failure of the red beam treated differently from the failure of another beam? ●

a. Because it carries more of the picture than the others
b. Because it produces more heat than the others
c. Because the information it carries must never be lost

155. Two screens have lost a beam and only one of them went monochrome, so what separates the two cases? ●

a. The monochrome one lost the beam that draws the green symbols
b. The monochrome one lost the beam that draws the warnings
c. The monochrome one lost two beams instead of one

156. What is wrong when a single scale on the primary flight display is replaced by a flag ●

a. The input signal for that parameter is not available
b. One electron beam of the display tube has failed
c. The display unit has started its overheat reaction

157. Several indications are gone and flags stand in their place, so what does this tell the engineer? ●

a. The display unit has begun to overheat
b. The display unit has an electron beam failure
c. The signals for those parameters are missing

158. One screen has lost some of its symbols and another shows flags in their place, so are the causes the same? ●

a. No, the flags come from a failed beam and the lost symbols from missing signals
b. Yes, both of them come from a failure inside the display unit
c. No, the flags come from missing signals and the lost symbols from a failed beam

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Answers — Display management and failures

135. a The indication disappears and a flag is shown instead. (L1, PDF p. 823)

136. b The first step is to switch off the raster colours. (L2, PDF p. 817)

137. b Only the strokes remain but none of the information is lost. (L2, PDF p. 817)

138. c Dropping the raster colours makes the tube produce less heat. (L2, PDF p. 817)

139. c The second step is to switch the display off altogether. (L2, PDF p. 817)

140. a The two causes given are an internal failure and problems in the cooling system. (L2, PDF p. 817)

141. a With one beam gone the picture contains no white at all. (L2, PDF p. 819)

142. a Information carried by the missing colour simply disappears from the screen. (L2, PDF p. 819)

143. a A failed red beam makes the whole display go monochrome. (L2, PDF p. 821)

144. a Only the failure of the red beam makes the display go monochrome. (L2, PDF p. 821)

145. b Only the indication belonging to the missing signal goes away. (L2, PDF p. 823)

146. a Engine and warning indications sit in the centre while the flight and navigation displays sit in front of the crew. (L3, PDF p. 811)

147. a The information of a failed primary flight display is switched automatically onto the navigation display. (L3, PDF p. 813)

148. a The takeover by another screen is automatic, so no crew action is needed to get the information back. (L3, PDF p. 813)

149. a Once it is back within the limits the unit activates its display again on its own. (L3, PDF p. 817)

150. c Losing the raster colours while keeping all the information is the first overheat step. (L3, PDF p. 817)

151. b The two step reaction goes on to its second step, in which the display is switched off. (L3, PDF p. 817)

152. c Loss of one beam removes both the white and everything drawn in that beam's colour. (L3, PDF p. 819)

153. a Warnings are drawn in red, so losing that beam would hide them unless the whole picture is made monochrome. (L3, PDF p. 821)

154. c The red beam draws the warning labels, which may not be allowed to disappear. (L3, PDF p. 821)

155. b Only the loss of the beam that draws the red warning labels turns the whole picture monochrome. (L3, PDF p. 821)

156. a A flag in place of one parameter is the sign that the input signal for it is missing. (L3, PDF p. 823)

157. c Flags are the signature of missing input signals, not of a fault inside the display unit. (L3, PDF p. 823)

158. c A flag marks a missing input, while symbols that vanish without a flag point to a beam failure in the tube. (L3, PDF p. 823)

Head up display and EFIS layout

PDF pages 825–839 · 17 questions

LEVEL 1 — RECALL

159. Which display units normally show the ND contents ●

a. The inboard display units
b. The outboard display units
c. The lower display unit

LEVEL 2 — UNDERSTAND AND EXPLAIN

160. Which movement of the pilot is avoided by the use of a head up display

a. Turning his head from the cockpit window to the overhead panel
b. Lowering his head from the cockpit window to the displays below
c. Lowering his head from the cockpit window to the control pedestal

161. Under which condition is the head up display meant to give guidance information on the ground and at take-off and landing ●

a. When the landing gear is extended
b. When the autopilot is engaged
c. When visibility is reduced

162. Which automatic procedure is the head up display used to monitor? ●

a. The auto approach
b. The auto land flare
c. The auto thrust reduction

LEVEL 3 — APPLY AND REASON

163. How does the head up display help the pilot at an airfield that has no instrument landing system ●

a. It gives the pilot information for the visual approach
b. It gives the pilot the localizer deviation
c. It gives the pilot the glide slope deviation

164. Visibility is poor on the approach, so why can the pilot keep his eyes on the outside world? ●

a. The information is presented on a screen at head height
b. The information is read out to him by a synthetic voice
c. The information is presented on a screen below the window

165. Why does the picture seen on the head up display change as the aircraft goes from take-off to cruise?

a. Because the display depends on the selected control panel
b. Because the display depends on the flight configuration
c. Because the display depends on the available input signals

166. Which segment of the head up display image is used while the automatic system flies the aircraft down to the runway

a. The ground roll information segment
b. The approach trajectory deviation segment
c. The system messages segment

167. Why can a parameter be absent from a display at one point in the flight and present at another

- **a.** Because the control panel is reset between the flight phases
- **b.** Because much data is only shown in the flight phase it belongs to
- **c.** Because the display units are tested one after the other in flight

168. A mechanic finds a glass cockpit aircraft with no separate DME or marker indicator fitted, so what does that tell him?

- **a.** Those two systems are not installed
- **b.** Those indications are handled by the EFIS
- **c.** Those indications are only on the standby panel

169. Which display is affected by four of the five basic EFIS control panel functions

- **a.** The lower display unit
- **b.** The ND
- **c.** The PFD

170. A pilot selects TCAS on the EFIS control panel, so where does the traffic picture appear?

- **a.** On the PFD
- **b.** On the ND
- **c.** On the standby instrument

171. A mechanic moves to a new aircraft type and finds an unfamiliar switch on its EFIS control panel, so how should he treat it?

- **a.** As a type specific additional function
- **b.** As a basic function common to every type
- **c.** As a function that acts on the standby instruments

172. An outboard display unit goes blank in flight, so what happens to the PFD picture? ●

- **a.** It can be switched to the inboard display unit
- **b.** It is moved to the standby attitude instrument
- **c.** It is lost until the display unit is replaced

173. In which two ways can the PFD reach the inboard display unit after an outboard display failure ●

- **a.** Automatically or manually
- **b.** Manually from either seat
- **c.** Automatically after a delay

174. A pilot wants his navigation picture on the lower display unit, so how does he get it there? ●

- **a.** He selects it on the source select module
- **b.** It appears there automatically
- **c.** He switches it there manually

175. Both outboard display units fail, so what does the crew lose on the inboard screens? ●

- **a.** The normal warning indications
- **b.** The normal navigation pictures
- **c.** The normal engine indications

Answers — Head up display and EFIS layout

159. a The two inboard display units normally carry the navigation display contents. (L1, PDF p. 839)

160. b The pilot no longer has to lower his head to the conventional displays below the window. (L2, PDF p. 825)

161. c That guidance is provided for the case of reduced visibility. (L2, PDF p. 825)

162. a One stated purpose is to monitor the auto approach. (L2, PDF p. 825)

163. a Where no instrument landing system exists the head up display supports the visual approach. (L3, PDF p. 825)

164. a Because the screen sits at head height he can read the guidance without looking down into the cockpit. (L3, PDF p. 825)

165. b The pilot is given different displays according to the flight configuration. (L3, PDF p. 827)

166. b Monitoring an automatic approach calls for the segment that shows how far the aircraft is off the trajectory. (L3, PDF p. 827)

167. b The flexibility of the screen lets much data be displayed only during the respective flight phase. (L3, PDF p. 831)

168. b The DME indicator and the marker lights are both among the instruments the EFIS replaces. (L3, PDF p. 831)

169. b Mode, range, TCAS and additional information selections all act on the ND, leaving only baro correction for the PFD. (L3, PDF p. 833)

170. b Selection of the TCAS indication is a basic function and the indication appears on the ND. (L3, PDF p. 833)

171. a Beyond the common basic functions, many types carry some type specific additional functions. (L3, PDF p. 833)

172. a On failure of the outboard display the PFD can be switched automatically or manually to the inboard DU. (L3, PDF p. 839)

173. a The switching to the inboard DU can be done automatically or manually. (L3, PDF p. 839)

174. c The captain's or F/O's ND can be switched onto the lower display unit manually. (L3, PDF p. 839)

175. b The inboard units normally show the ND, so taking the PFDs onto them costs the crew those navigation pictures. (L3, PDF p. 839)

EFIS displays and symbols

PDF pages 841–851 · 24 questions

LEVEL 1 — RECALL

176. On which display unit is the PFD normally shown ●

a. The outboard one
b. The inboard pair
c. The lower one

177. From which viewpoint does the navigation display present the aircraft situation ●

a. From behind
b. From above
c. From the side

178. Which conventional instrument is fully replaced by the navigation display ●

a. The HSI
b. The ADI
c. The RMI

179. During which part of the flight does the ND ILS mode indicate its parameters ●

a. During the ILS approach
b. During the initial climb out
c. During the taxi out

180. A pilot selects the ILS mode on his ND, so which two deviations does he then see? ●

a. Localizer and track
b. Glideslope and vertical speed
c. Localizer and glideslope

181. Which three items are added in the navigation sector mode of the ND ●

a. The waypoint, the range selection and the route indication
b. The range selection, the localizer scale and the waypoint
c. The waypoint, the glideslope scale and the route indication

LEVEL 2 — UNDERSTAND AND EXPLAIN

182. A pilot's PFD appears on the inboard screen in flight, so what has most likely happened? ●

a. The normal display unit has failed
b. The control panel has been reset
c. The baro correction has been changed

183. What is the small aircraft symbol on the navigation display there for ●

a. To show the attitude of the aircraft
b. To show the aircraft type being flown
c. To act as a reference for the picture

184. Which navigation point does the basic waypoint data on the ND refer to ●

a. The destination one
b. The next one
c. The last one passed

185. A trainee asks what ground speed on the navigation display means, so what is the answer? ●

a. The speed limit on the ground
b. The speed through the air
c. The speed above the ground

186. Which ND parameter comes from somewhere other than the navigation systems ●

a. The wind velocity
b. The heading
c. The ground speed

187. Where on the navigation display is the small aircraft symbol normally shown?

a. At the edge of the selected range
b. At the top of the compass rose
c. In the centre of the screen

188. Which aircraft system supplies the parameters shown in the ND ILS mode ●

a. The air data computer system
b. The flight management computer system
c. The instrument landing system

189. A route indication is missing from a navigation display in the sector mode, so which system should be suspected? ●

a. The instrument landing system
b. The ground proximity warning system
c. The flight management system

LEVEL 3 — APPLY AND REASON

190. Which system supplies the AFCS information shown on the PFD

a. The navigation system
b. The air data system
c. The autoflight system

191. The pressure setting on a PFD will not change, so which control should be suspected first?

a. The baro correction on the EFIS control panel
b. The mode selection on the EFIS control panel
c. The range selection on the EFIS control panel

192. Which two deviations are shown together on the PFD in the ILS mode

a. The localizer and the glide slope
b. The glide slope and the radio height
c. The localizer and the vertical speed

193. A mechanic is asked why there is no separate ILS indicator in an EFIS cockpit, so what should he say?

a. The ILS is only monitored from the standby instrument
b. The PFD takes that job in the ILS mode
c. The ILS picture is only available on the ground

194. Why does the wind indication stay on the navigation display whatever mode is selected ●

a. Because the control panel cannot reach that part of the screen
b. Because it is one of the basic items always present
c. Because it is only drawn by the air data computer

195. A pilot changes the ND mode and loses an indication he wanted, so what does that tell him about it? ●

a. It is not one of the basic always present items
b. It is only shown while the autopilot is engaged
c. It is only driven by the flight management system

196. The navigation systems feeding an ND are lost but the heading is still drawn, so what does that say about the heading? ●

a. It is the one item that is not navigation data
b. It is navigation data like every one of the others
c. It is drawn by the flight management computer

197. Why can the localizer deviation be read on the PFD and on the ND at the same time ●

a. Because the ND copies the whole PFD picture in that mode
b. Because the ILS feeds only the navigation display units
c. Because both displays can present ILS parameters

198. Why does selecting the navigation sector mode change what the pilot can read on the display ●

a. Because that mode removes the compass rose from the picture
b. Because that mode moves the picture onto the lower display unit
c. Because that mode adds flight management data

199. A pilot swaps from the ILS mode to the navigation sector mode, so what replaces the ILS parameters on his display? ●

a. Inertial reference information
b. Flight management information
c. Radio altimeter information

Answers — EFIS displays and symbols

176. a The PFD is normally presented on the outboard display unit. (L1, PDF p. 841)

177. b The ND shows the situation of the aircraft seen from above. (L1, PDF p. 847)

178. a The ND fully replaces the horizontal situation indicator. (L1, PDF p. 847)

179. a The ILS mode indicates ILS parameters during the ILS approach. (L1, PDF p. 849)

180. c The ILS mode of the ND indicates localizer and glideslope deviation. (L1, PDF p. 849)

181. a The nav sector mode additionally displays waypoint, range selection and route indication. (L1, PDF p. 851)

182. a The PFD is switched to the inboard display unit on failure of its normal one. (L2, PDF p. 841)

183. c The small aircraft symbol is used as a reference and is normally in the centre of the screen. (L2, PDF p. 847)

184. b The waypoint data gives information about the next navigation point. (L2, PDF p. 847)

185. c Ground speed is the speed above the ground. (L2, PDF p. 847)

186. b Apart from the heading, all information on the ND is based on data from the navigation systems. (L2, PDF p. 847)

187. c The small aircraft symbol is normally displayed in the centre of the screen, while the top of the compass rose is where the actual heading is read. (L2, PDF p. 847)

188. c The ILS mode indicates parameters coming from the instrument landing system. (L2, PDF p. 849)

189. c The route indication in that mode is supplied by the flight management system. (L2, PDF p. 851)

190. c The AFCS information on the PFD comes from the autoflight system. (L3, PDF p. 843)

191. a The pressure setting on the PFD is the baro correction, which is set from the EFIS control panel. (L3, PDF p. 843)

192. a The ILS mode displays the localizer and the glide slope alongside the ILS source and the marker. (L3, PDF p. 845)

193. b The PFD itself is used as the display for the instrument landing system. (L3, PDF p. 845)

194. b Wind direction and velocity is one of the basic pieces of information present no matter what is selected. (L3, PDF p. 847)

195. a Only the listed basic pieces of information are present no matter what is selected on the control panel. (L3, PDF p. 847)

196. a Apart from the heading, all information on the ND is based on data from the navigation systems. (L3, PDF p. 847)

197. c The PFD has an ILS mode and the ND has an ILS mode, and both show the localizer indication. (L3, PDF p. 849)

198. c The nav sector mode adds the waypoint, range selection and route indication, all from the FMS. (L3, PDF p. 851)

199. b The ILS mode shows ILS parameters while the nav sector mode adds information from the flight management system. (L3, PDF p. 851)

Central warning systems

PDF pages 859–871 · 58 questions

LEVEL 1 — RECALL

200. How do the pushbuttons appear while everything is operating normally ●

a. Amber
b. White
c. Dark

201. Which colour is used when an immediate action is necessary ●

a. Red
b. Amber
c. White

202. Which colour indicates the normal operation of a backup system ●

a. Green
b. Blue
c. White

203. Which colour indicates the normal operation of a temporarily used system ●

a. Green
b. Amber
c. Blue

204. Where is the centralised caution panel installed ●

a. On the centre pedestal
b. At the pilots' eye level
c. On the overhead panel

205. Where is the local warning light of a system covered by the centralised caution panel fitted ●

a. On the glareshield
b. On the centralised caution panel
c. On the overhead panel

206. What two forms of information do central warning displays give the flight crew ●

a. Text and aural information
b. Graphic and tactile information
c. Text and graphic information

207. What kind of system is ECAM ●

a. A flight management system
b. A central warning system
c. An autoflight system

208. Which two displays make up ECAM in the cockpit ●

a. The primary flight display and the system display
b. The engine/warning display and the navigation display
c. The engine/warning display and the system display

209. Apart from calling up pages and messages, which display property is set from the ECAM control panel ●

a. The contrast ratio
b. The brightness
c. The refresh rate

210. What is simulated when the TO CONFIG pushbutton is used ●

a. The take-off power configuration
b. The approach configuration
c. The emergency electrical configuration

211. Which two kinds of page does the SD of an ECAM installation show ●

a. The engine and warning pages
b. The system and status pages
c. The memo and alert pages

212. How many system pages are provided in total ●

a. Five
b. Eight
c. Twelve

LEVEL 2 — UNDERSTAND AND EXPLAIN

213. What happens to the related system when an illuminated pushbutton is pressed

a. Its switching state is changed
b. Its warning is only acknowledged
c. Its light test is only started

214. A pushbutton on the overhead panel has just lit up, so what does that tell the crew? ●

a. The pushbutton lamp is being tested by the crew
b. A local warning or an abnormal switching state exists
c. The system has returned to its normal switching state

215. Why are the pushbutton lights given a colour code ●

a. To make abnormal conditions easy to identify
b. To show which bus supplies each panel
c. To group the pushbuttons by aircraft system

216. Besides an abnormal pushbutton position, which condition is shown by a white light ●

a. A maintenance operation
b. A backup system running
c. A failure needing awareness

217. Which shortcoming of the master caution and master warning lights led to a centralised panel ●

a. They cannot be cancelled once they are on
b. They do not stay on long enough to be seen
c. They do not say which system is affected

218. What does the centralised caution panel tell the crew about a fault ●

a. How many flight hours remain before rectification
b. Which circuit breaker has tripped
c. Which particular system it is in

219. How is the master caution light cancelled ●

a. By pressing the local warning light
b. By switching off the affected system
c. By pressing its light cover

220. Why does the crew cancel the master caution light rather than leave it on? ●

a. So that another fault can be indicated
b. So that the local warning light goes out
c. So that the centralised panel can be tested

221. What does the information on a central warning display describe ●

a. The flight phase being computed
b. The present failure condition
c. The maintenance action already carried out

222. An engineer compares the central warning systems of two different aircraft types, so what will he mostly find?

a. Different functions under the same name
b. The same function under different names
c. The same name and the same hardware

223. Under what condition is the symbol ADV displayed ●

a. When a status page becomes available
b. When ECAM is working as a mono display
c. When a caution message is cancelled

224. One of the two ECAM displays has failed, so which symbol appears in the lower part of the engine/warning display? ●

a. STS
b. RCL
c. ADV

225. Which two pages are combined in the cruise page ●

a. The air and hydraulic pages
b. The engine and air pages
c. The engine and fuel pages

226. A memo message has to be read, so which display and which part of it does the crew use? ●

a. The lower part of the system display
b. The upper part of the engine/warning display
c. The lower part of the engine/warning display

227. Which three things can be done to warnings and messages from the ECAM control panel ●

a. Record, print or transmit them
b. Cancel, delete or call them up again
c. Dim, enlarge or colour them

228. In which position must the slats and flaps be for the take-off configuration check to pass ●

a. In the T/O range
b. In the fully extended position
c. In the fully retracted position

229. What condition of the speed brakes is required by the take-off configuration check ●

a. They must be extended
b. They must be retracted
c. They must be armed

230. What condition of the wheel brakes is required by the take-off configuration check ●

a. They must be released
b. They must not be overheated
c. They must be parked

231. The take-off configuration check is run with a cabin door still open, so what will the crew find? ●

a. The check will be bypassed automatically
b. The normal message will not appear
c. The master caution will be inhibited

232. A continuous aural alert is sounding and the crew wants it silenced, so which ECAM control panel pushbutton is used? ●

a. CLR
b. RCL
c. EMER CANC

233. What happens when the ALL pushbutton is pressed and held ●

a. The status page is called up again after five seconds
b. The system pages are frozen on the display
c. The system pages are called up at one second intervals

234. What happens when the ALL pushbutton is pressed only briefly ●

a. The displayed system page is deleted
b. The system page moves forward by one page
c. The system pages cycle continuously

235. When does the light in the CLR pushbutton come on ●

a. When the recall function has just been used
b. When warning, caution or status messages appear
c. When the system pages are being cycled by the crew

236. Which message is removed each time the CLR pushbutton is pushed ●

a. Every warning at once
b. The warning at the top
c. The oldest warning in the list

237. What reappears on the display after the last clearance has been made with CLR ●

a. The memo messages
b. The status page
c. The caution messages

238. How long after the status pushbutton is used do the normal messages return when no status messages are present ●

a. After 10 seconds
b. After 5 seconds
c. After 2 seconds

239. The brightness of the engine/warning and system displays has to be changed, so which panel is used? ●

a. The ECAM control panel
b. The EICAS display select panel
c. The centralised caution panel

240. How many system pages can be shown on the system display at any one time ●

a. Four
b. One
c. Two

241. Which two air temperatures are shown among the permanent data on a system page ●

a. TAT and EGT
b. SAT and ITT
c. TAT and SAT

242. Which item is shown as an alternative to the G-load among the permanent data ●

a. The static air temperature
b. The selected altitude
c. The gross weight

243. A pilot needs the total air temperature while a system page is up, so where is it shown? ●

a. On the upper part of the page
b. On the lower part of the page
c. On the engine/warning display

LEVEL 3 — APPLY AND REASON

244. A system is working normally but its pushbutton has been left out of its normal switching position, so what will the crew see on that pushbutton? ●

a. It will stay dark because no fault exists
b. It will be lit even though no fault exists
c. It will flash until the fault is cleared

245. A pushbutton carries an amber FAULT legend, so how urgent is the crew action? ●

a. Action is not needed at all
b. Action is needed at once
c. Action is needed but not at once

246. A pushbutton is lit in a colour that reports normal operation, so which two colours could it be? ●

a. Green or blue
b. Red or amber
c. Amber or white

247. A system has been selected off and no fault exists, so which colour will its pushbutton show? ●

a. Amber
b. Green
c. White

248. Which two colours on a pushbutton both report a failure of some kind ●

a. White and green
b. Red and amber
c. Green and blue

249. A fault occurs in a system whose local light is on the overhead panel out of the pilot's sight, so how does the crew learn which system it is? ●

a. The overhead local light is repeated on the glareshield
b. The master caution light names the affected system
c. The centralised caution panel names the affected system

250. A pilot sees STS in the lower part of the engine/warning display but no status page is being shown, so where will that status page appear once it is called up? ●

a. On the system display
b. On the engine/warning display
c. On the centralised caution panel

251. A page made up of the engine and air pages is on the system display, so where is the aircraft in its flight? ●

a. In the take-off run
b. In the cruise
c. In the approach

252. A crew member has cleared several caution messages with CLR and then wants to read them again, so what must be done? ●

a. Press RCL to recall them
b. Press CLR again to restore them
c. Press STS to bring up the status page

253. The status pushbutton has been pressed and after a short time the normal display returns on its own, so what does that tell the crew? ●

a. The status page has been cleared by CLR
b. The ECAM is working as a mono display
c. No status messages are present

254. A level 3 alert has triggered an aural warning and the master warning lights, and the crew uses EMER CANC, so what is left showing on the ECAM? ●

a. Nothing at all is left on the display
b. Only the system page is cancelled
c. The message on the display stays

255. STS is showing on the engine/warning display and the crew presses the status pushbutton, so what appears? ●

a. The normal messages, since no status messages exist
b. The status page with its messages
c. The cruise page with its engine data

256. A pilot wants to see a second system page while one is already up, so what happens to the page already displayed? ●

a. It is replaced, because only one can be shown at a time
b. It stays, because two pages can be shown side by side
c. It stays, because the second page opens on the E/WD

257. A pilot reading a system page wants the gross weight, so where on that page does he look? ●

a. On the right side, among the inoperative systems
b. On the upper part, among the engine parameters
c. On the lower part, among the permanent data

Answers — Central warning systems

200. c All pushbuttons are dark during normal operation. (L1, PDF p. 859)

201. a Red is the colour used when an immediate action is necessary. (L1, PDF p. 859)

202. a Green indicates the normal operation of a backup system. (L1, PDF p. 860)

203. c Blue indicates the normal operation of a temporarily used system. (L1, PDF p. 860)

204. b The centralised caution panel is installed at the pilots' eye level. (L1, PDF p. 861)

205. c The local warning light sits on the overhead panel while the centralised panel sits in view. (L1, PDF p. 861)

206. c The displays provide text and graphic information. (L1, PDF p. 863)

207. b ECAM is a central warning system. (L1, PDF p. 865)

208. c ECAM is present as the E/WD and the SD. (L1, PDF p. 865)

209. b The control panel is used to control the brightness of the ECAM display units. (L1, PDF p. 867)

210. a The TO CONFIG pushbutton simulates the T/O power configuration. (L1, PDF p. 867)

211. b The SD shows the system and the status pages. (L1, PDF p. 871)

212. c One of twelve system pages can be shown. (L1, PDF p. 871)

213. a Pressing the illuminated pushbutton changes the state of the switch behind it. (L2, PDF p. 859)

214. b A pushbutton lights only for a local warning or for an abnormal switching state. (L2, PDF p. 859)

215. a The colour code exists to help in the easy identification of abnormal conditions. (L2, PDF p. 859)

216. a White covers a pushbutton in an abnormal position or a maintenance operation. (L2, PDF p. 860)

217. c The master lights give no information about which system is affected or which local light is on. (L2, PDF p. 861)

218. c The panel indicates a fault in a particular system. (L2, PDF p. 861)

219. c The master caution light is cancelled by pressing its light cover. (L2, PDF p. 861)

220. a Cancelling the light frees it so that a further fault can be indicated. (L2, PDF p. 861)

221. b The displays describe the present failure condition. (L2, PDF p. 863)

222. b Central warning systems in different types generally share the function but not the name. (L2, PDF p. 863)

223. b ADV appears when ECAM is working as a mono display, that is with only one display. (L2, PDF p. 865)

224. c With only one ECAM display left, ECAM works as a mono display and ADV is shown. (L2, PDF p. 865)

225. b The cruise page consists of the engine and air pages. (L2, PDF p. 865)

226. c Memo messages are in the lower part of the E/WD. (L2, PDF p. 865)

227. b The panel cancels, deletes or calls up again the warnings and messages. (L2, PDF p. 867)

228. a The check requires the slats and flaps to be in the T/O range. (L2, PDF p. 867)

229. b The check requires the speed brakes to be retracted. (L2, PDF p. 867)

230. b The check requires the wheel brakes not to be OVHT. (L2, PDF p. 867)

231. b Door closed is one of the items checked, so an open door prevents the normal result. (L2, PDF p. 867)

232. c The EMER CANC pushbutton cancels the aural warnings. (L2, PDF p. 867)

233. c Pressing and holding ALL calls up the system pages at one second intervals. (L2, PDF p. 867)

234. b A brief press moves the system page forward by one page. (L2, PDF p. 867)

235. b The CLR light comes on when warning, caution or status messages appear on the ECAM display. (L2, PDF p. 867)

236. b Each push of CLR deletes the warning at the top. (L2, PDF p. 867)

237. a After the last clearance the memo messages reappear. (L2, PDF p. 867)

238. b With no status messages present the normal messages are displayed again after 5 seconds. (L2, PDF p. 867)

239. a Brightness is controlled from the ECAM control panel. (L2, PDF p. 867)

240. b Only one of the twelve system pages can be displayed at any one time. (L2, PDF p. 871)

241. c The permanent data includes TAT and SAT. (L2, PDF p. 871)

242. b The permanent data ends with G-load or selected altitude. (L2, PDF p. 871)

243. b TAT is one of the permanent data items on the lower part of the system page. (L2, PDF p. 871)

244. b A pushbutton lights for an abnormal switching state as well as for a local warning, so it is lit with no fault present. (L3, PDF p. 859)

245. c Amber is used when action is required but not necessarily at once. (L3, PDF p. 859)

246. a Green and blue are the two colours for normal operation, green for a backup system and blue for a temporarily used system. (L3, PDF p. 860)

247. c A pushbutton left in an abnormal position is coded white, and no failure is being reported. (L3, PDF p. 860)

248. b Red and amber are the two colours for a failure, red needing immediate action and amber only awareness. (L3, PDF p. 860)

249. c The master light says only that something is wrong, so the panel at eye level is what identifies the system. (L3, PDF p. 861)

250. a STS says a status page is available and the status page is displayed on the SD. (L3, PDF p. 865)

251. b The cruise page, which consists of the engine and air pages, is the one shown in cruise flight. (L3, PDF p. 865)

252. a CLR deletes the messages one by one and RCL is the pushbutton that recalls them. (L3, PDF p. 867)

253. c The normal messages come back by themselves only when there are no status messages to show. (L3, PDF p. 867)

254. c EMER CANC removes the aural warnings, the master warning lights and the caution messages, so a warning message itself is not in that list. (L3, PDF p. 867)

255. b STS means a status page is available, so pressing the pushbutton brings that page up rather than returning the normal messages. (L3, PDF p. 867)

256. a Only one of the twelve system pages can be displayed at any one time. (L3, PDF p. 871)

257. c GW is among the permanent data on the lower part of the system page. (L3, PDF p. 871)

Alert levels, colours and flight phases

PDF pages 877–893 · 49 questions

LEVEL 1 — RECALL

258. Which aircraft condition is associated with a level 3 alert ●

a. A configuration to monitor
b. An abnormal configuration
c. An emergency configuration

259. Which aircraft condition is associated with a level 2 alert ●

a. An emergency configuration
b. A configuration to monitor
c. An abnormal configuration

260. What priority do level 3 warnings carry ●

a. The highest priority
b. A medium priority
c. The lowest priority

261. How quickly must the crew react to a level 3 warning ●

a. Without any delay
b. Before the next flight phase
c. Before the next scheduled check

262. What influence does a level 2 caution have on the flight ●

a. An influence only during the approach
b. No direct influence
c. A direct and immediate influence

263. What is required of the crew when a level 2 caution appears ●

a. Greater attention
b. An immediate reaction
c. No reaction at all

264. On which display do level 1 cautions appear ●

a. The centralised caution panel
b. The engine/warning display
c. The system display

265. Which indications are absent for a level 1 caution ●

a. Memo messages and status messages
b. Aural alarms and warning lights
c. Text messages and system pages

266. Which flight phase does the flight warning system select by default when it has no data ●

a. The preflight phase
b. The approach phase
c. The cruise phase

267. Which parameters are permanently displayed on the upper EICAS ●

a. The status information
b. The secondary engine parameters
c. The primary engine parameters

268. How many EFIS/EICAS interface units are fitted ●

a. Four
b. Three
c. Two

269. Which two parameters are the primary engine parameters on the EICAS display ●

a. N2 rotor speed and fuel flow in kilogrammes
b. N1 rotor speed and exhaust gas temperature
c. Oil pressure and oil temperature in degrees

270. Which colour is used for a Level A warning message in the alert message field ●

a. Red
b. White
c. Amber

271. How many manual snapshots can be stored for each system ●

a. Up to 5
b. Up to 10
c. Up to 3

272. A snapshot has just been taken because a parameter went outside its preset value, so where is it kept? ●

a. In the manual event non-volatile memory
b. In the auto event non-volatile memory
c. In the real time data storage buffer

LEVEL 2 — UNDERSTAND AND EXPLAIN

273. Which alert level is classified as a warning rather than a caution ●

a. Level 2
b. Level 3
c. Level 1

274. In what manner does the master warning light behave for a level 3 warning ●

a. It flashes
b. It stays dark
c. It stays steady

275. Besides the level 3 warning itself, which page may be shown if one is available ●

a. A system page, if one is available
b. A cruise page, if one is available
c. A status page, if one is available

276. How many chimes are sounded for a level 2 caution ●

a. None
b. Three
c. One

277. What kind of system does a level 1 caution draw attention to ●

a. A serviceable system temporarily switched off
b. A defective system needed for the flight to continue
c. A defective system not required for the flight to continue

278. A caution message has appeared on the engine/warning display with no sound and no light, so which level has been reached? ●

a. Level 3
b. Level 1
c. Level 2

279. From which system does the flight warning system take its compressed or not compressed signal

a. The air data system
b. The brake control system
c. The landing gear system

280. Which number is given to the default flight phase of the flight warning system ●

a. 12
b. 8
c. 1

281. Which four items can be shown on the lower EICAS ●

a. Primary engine parameters, synoptics, memo messages and maintenance pages
b. Secondary engine parameters, system pages, status information and alert messages
c. Secondary engine parameters, synoptics, status information and maintenance pages

282. Which panel brings synoptics and status information onto the lower screen ●

a. The ECAM control panel
b. The EICAS display select panel
c. The centralised caution panel

283. Through which unit are the maintenance pages called up on the lower EICAS ●

a. The EDSP
b. The EIU
c. The CMC

284. Which data appears on the lower EICAS immediately after power-up ●

a. The primary engine data
b. The status information
c. The secondary engine data

285. Which data bus supplies the EICAS display units ●

a. An ARINC 629 data bus
b. An ARINC 573 data bus
c. An ARINC 429 data bus

286. A message shows in the alert message field in amber and indented, so which level is it? ●

a. Level B caution
b. Level D memo
c. Level C advisory

287. Why is information that is not required removed from the EICAS display ●

a. To reduce the heat produced by the screens
b. To save electrical power in the display units
c. To maintain a quiet, dark flight deck

288. What makes an automatic snapshot occur for a system ●

a. Parameters that exceed preset values
b. Selection of the real time display mode
c. A request made by the maintenance crew

289. What causes a manual snapshot of the real time format to be taken ●

a. An action by the flight or maintenance crew
b. A parameter that goes out of its limits
c. A report sent to the printer or to ACARS

LEVEL 3 — APPLY AND REASON

290. An engine fire breaks out in flight, so what must the crew do about it? ●

a. Note it and continue, as the flight is not affected
b. React before the next flight phase begins
c. React immediately, as it is a real emergency

291. A level 3 warning and a level 1 caution are annunciated at the same moment, so which one is handled first? ●

a. The caution, which has the highest priority
b. The warning, which has the highest priority
c. Either one, since the two carry the same priority

292. An instructor asks what separates a warning from a caution in practice, so what is the difference in the reaction required? ●

a. A warning needs an immediate reaction, a caution does not
b. A caution needs an immediate reaction, a warning does not
c. Each one needs the same immediate reaction from the crew

293. A single chime sounds and the master caution light comes on, so what does the crew know about the failure? ●

a. It is a real emergency needing an immediate reaction
b. It has no direct influence on the flight
c. It gives no aural alarm and no warning light

294. A failure needs greater attention but no immediate reaction, so which aural alert will the crew hear? ●

a. A single chime
b. No sound at all
c. A continuous alarm

295. A system that the flight does not need has become defective, so what will the crew hear? ●

a. A continuous alarm from the master warning
b. Nothing, as there is no aural alarm
c. A single chime from the master caution

296. A recorder fault and a stall warning occur in the same flight, so how do the two alerts differ in what the crew gets? ●

a. Each one gives an aural alarm but only one gives a light
b. One gives a single chime while the other gives no sound
c. One gives no sound or light while the other gives both

297. Which indication may be given at all three alert levels if one is available ●

a. A light coming on at the eye level
b. A system page on the system display
c. A chime sounding from the loudspeaker

298. A technician wants to bring the maintenance pages up on the lower EICAS, so why will the display select panel not do it? ●

a. Because that panel only works after the aircraft is powered up
b. Because that panel only selects synoptics and status information
c. Because that panel only selects the primary engine parameters

299. An aircraft has just been powered up and no selection has been made, so what is seen on the two EICAS display units? ●

a. Primary engine data above and secondary engine data below
b. Primary engine data above and the status information below
c. Secondary engine data above and primary engine data below

300. All three interface units have failed, so what will the EICAS display units show? ●

a. Nothing, because all the data to the screens passes through them
b. Primary engine data only, taken direct from the engines
c. Maintenance pages only, taken direct from the CMC

301. A pilot needs the secondary engine parameters while a synoptic is up on the lower EICAS, so what must happen on that display? ●

a. The synoptic and the parameters appear together
b. The upper display must be used to show them
c. The synoptic must be deselected to show them

302. A synoptic is wanted on the lower EICAS straight after power-up, so what must be done? ●

a. It must be called up through the maintenance computer
b. It must be selected on the display select panel
c. It is already there after power-up

303. The upper EICAS display unit is showing data that cannot be deselected, so what is on it? ●

a. The secondary engine parameters
b. The synoptics of the systems
c. The primary engine parameters

304. Every indication that is not needed has been removed from the main EICAS display, so which two groups are still shown? ●

a. The primary engine parameters and the fuel system data
b. The cabin altitude and the duct pressure readouts
c. The landing gear status and the flap indications

305. A caution and an advisory both show in amber in the alert message field, so how is the advisory told apart? ●

- **a.** It is flashing
- **b.** It is underlined
- **c.** It is indented

306. An engine exceedance occurs while the aeroplane stands with its parking brake set, so what becomes of the event? ●

- **a.** It is lost, since the brake has to be released first
- **b.** It is lost, since two engines have to be running
- **c.** It is stored, since that storage is not inhibited

Answers — Alert levels, colours and flight phases

258. c Level 3 is the warning raised for an emergency configuration. (L1, PDF p. 878)

259. c Level 2 is the caution raised for an abnormal configuration. (L1, PDF p. 878)

260. a Level 3 warnings are the warnings with the highest priority. (L1, PDF p. 879)

261. a A level 3 warning is a real emergency and the crew must react immediately. (L1, PDF p. 879)

262. b Level 2 cautions have no direct influence on the flight. (L1, PDF p. 881)

263. a A level 2 caution calls for greater attention. (L1, PDF p. 881)

264. b Level 1 cautions appear on the E/WD. (L1, PDF p. 883)

265. b There are no aural alarms or warning lights for a level 1 caution. (L1, PDF p. 883)

266. c With the data unavailable the FWS defaults to the cruise phase. (L1, PDF p. 885)

267. c The upper EICAS permanently displays the primary engine parameters. (L1, PDF p. 887)

268. b There are three EFIS/EICAS interface units. (L1, PDF p. 887)

269. b The primary engine parameters are the N1 rotor speed and the exhaust gas temperature. (L1, PDF p. 889)

270. a Level A warning messages are shown in red. (L1, PDF p. 889)

271. a Storage is available for up to 5 manual snapshots per system. (L1, PDF p. 893)

272. b Auto snapshots go to the auto event non-volatile memory. (L1, PDF p. 893)

273. b Level 3 carries the title warnings while levels 2 and 1 are cautions. (L2, PDF p. 877)

274. a The master warning light flashes for a level 3 warning. (L2, PDF p. 879)

275. a If available, a system page may also appear on the SD. (L2, PDF p. 879)

276. c A level 2 caution is given with a single chime. (L2, PDF p. 881)

277. c Level 1 cautions point at a defective system which is not required for the flight to continue. (L2, PDF p. 883)

278. b Level 1 is the only level with no aural alarm and no warning light. (L2, PDF p. 883)

279. c The gear compressed or not compressed information comes from the L/G system. (L2, PDF p. 885)

280. b The default is the cruise phase, numbered 8. (L2, PDF p. 885)

281. c The lower EICAS can show secondary engine parameters, synoptics, status information and maintenance pages. (L2, PDF p. 887)

282. b Synoptics and status information are selected with the EICAS display select panel (EDSP). (L2, PDF p. 887)

283. c The maintenance pages are called up via the CMC. (L2, PDF p. 887)

284. c After power-up the lower EICAS shows the secondary engine data. (L2, PDF p. 887)

285. c The display units are supplied via an ARINC 429 data bus. (L2, PDF p. 887)

286. c Only the Level C advisory messages are both amber and indented. (L2, PDF p. 889)

287. c Removing unneeded information is what keeps the flight deck quiet and dark. (L2, PDF p. 889)

288. a An auto snapshot is taken when parameters go beyond their preset values. (L2, PDF p. 893)

289. a A manual snapshot follows an action by the flight or maintenance crew. (L2, PDF p. 893)

290. c Engine fire is a level 3 warning and a level 3 warning is a real emergency needing an immediate reaction. (L3, PDF p. 879)

291. b Level 3 warnings are the warnings with the highest priority. (L3, PDF p. 879)

292. a The level 3 warning is a real emergency needing an immediate reaction, while the level 2 caution has no direct influence on the flight. (L3, PDF p. 879)

293. b A single chime with the master caution light is a level 2, and level 2 failures have no direct influence on the flight. (L3, PDF p. 881)

294. a Greater attention without an immediate reaction is a level 2, and a level 2 is given with a single chime. (L3, PDF p. 881)

295. b A defective system not required to continue the flight is a level 1 caution, and level 1 has no aural alarm. (L3, PDF p. 883)

296. c A recorder fault is a level 1 with no aural alarm or warning light, while a stall warning is a level 3 with an aural alarm and a flashing master warning light. (L3, PDF p. 883)

297. b Each of the three levels carries the note that a system page may also appear on the SD if available. (L3, PDF p. 883)

298. b The EDSP selects synoptics and status information while the maintenance pages come via the CMC. (L3, PDF p. 887)

299. a After power-up the upper unit shows the primary and the lower unit the secondary engine data. (L3, PDF p. 887)

300. a The display units are fed by one of the three EIUs, and all the important systems supply their data to those units. (L3, PDF p. 887)

301. c The lower EICAS carries the secondary parameters, synoptics, status and maintenance pages in turn, while the upper one is permanently the primary parameters. (L3, PDF p. 887)

302. b After power-up the lower unit shows the secondary engine data, so a synoptic has to be selected on the EDSP. (L3, PDF p. 887)

303. c The upper EICAS permanently displays the primary engine parameters, the selectable items being on the lower unit. (L3, PDF p. 887)

304. a Those two groups are the ones stated to show at all times. (L3, PDF p. 889)

305. c The advisory is the amber message that is indented. (L3, PDF p. 889)

306. c Engine exceedance page storage is not inhibited, so the running and brake conditions do not apply to it. (L3, PDF p. 893)

Ground proximity warning principles

PDF pages 897–905 · 31 questions

LEVEL 1 — RECALL

307. What is the purpose of the GPWS ●

a. To warn the flight crew when the aircraft comes too close to the ground
b. To warn the flight crew of another aircraft on a conflicting track
c. To warn the flight crew of a windshear encountered on approach

308. By which two means is the basic GPWS warning given to the pilot ●

a. A master caution and a single chime
b. An aural alert and annunciator lights
c. A flashing display and a stick shaker

309. On which display does the enhanced ground proximity warning system show the surrounding terrain ●

a. On the navigation display
b. On the primary flight display
c. On the engine indication display

310. Which three parts are the main parts of a ground proximity warning system ●

a. One warning computer, a radio altimeter and a speaker
b. One warning computer, indications and inhibit switches
c. Two warning computers, indications and a control panel

311. A basic GPWS is checked, so at which radio altitude does its operative range begin? ●

a. 50 ft
b. 30 ft
c. 10 ft

312. An aircraft is fitted with the enhanced system, so which two functions does it have on top of the basic modes? ●

a. Terrain awareness alerting and display, and terrain clearance floor
b. Radio altitude callouts, and the modulation of alerting envelopes
c. Windshear alerting for take-off, and excessive bank angle alerting

313. The enhanced computer needs the height above the ground for its warning calculation, so which unit supplies it? ●

a. The instrument landing system
b. The digital air data computer
c. The downward looking radio altimeter

LEVEL 2 — UNDERSTAND AND EXPLAIN

314. A ground proximity alert sounds in the flight deck, so what kind of voice is heard? ●

a. A synthetic voice
b. A rising tone sequence
c. A recorded crew voice

315. One class of aircraft only rarely carries a ground proximity warning system, so which class is it? ●

a. Four-engined aeroplanes
b. Twin-engined aeroplanes
c. Helicopters

316. Where else are GPWS alert messages presented in some aircraft ●

a. On the engine indication display
b. On the navigation displays
c. On the primary flight displays

317. In which aeroplanes is the ground proximity warning system installed ●

a. In the very small ones only
b. In all of them without any exception
c. In all of them except the very small ones

318. Which signal can the terrain functions of the enhanced system do without ●

a. The glide slope deviation
b. The radio altitude
c. The barometric altitude

319. A crew is told their fit is the advanced version, so what does it do that the basic one does not? ●

a. It shows the terrain around the aircraft
b. It lights annunciators in front of the pilot
c. It sounds its alerts with a synthetic voice

320. From which sources does the GPWC receive its inputs ●

a. From the autopilot and the flight control computers
b. From the flight data recorder and the engine computers
c. From an air data computer and various navigation systems

321. Only some aeroplanes carry one of the three main parts of the system, so which part is it? ●

a. The inhibit switches
b. The warning computer
c. The indications

322. Which kind of accident is the enhanced system intended to prevent ●

a. Controlled flight into terrain
b. Loss of control in flight
c. Runway incursion on landing

323. With what does the enhanced system compare the aircraft position ●

a. With an internal database
b. With the weather radar picture
c. With the radio altitude signal

324. Which two sources can supply the aircraft position used in that comparison ●

a. The FMS or the GPS
b. The ILS or the VOR
c. The ADC or the IRS

325. Windshear alerting is fitted to an aircraft, so which mode number does it carry?

a. Mode 6
b. Mode 7
c. Mode 5

326. What does the enhanced system do with the variety of aircraft parameters it accepts ●

a. It applies alerting algorithms to them
b. It transmits them to the ground station
c. It records them for later readout

327. Which two products were introduced in the late 1980s

a. MARK II for analogue aircraft and MARK III for digital aircraft
b. MARK V for digital aircraft and MARK VII for analogue aircraft
c. MARK VI for digital aircraft and MARK I for analogue aircraft

328. Which two introductions improve the terrain awareness and the warning times of the enhanced system ●

a. The terrain display and the terrain data base look ahead protection
b. The altitude callouts and the alert for an excessive bank angle
c. The terrain clearance floor and the modulation of the envelopes

329. Why can the warning time be short when the aircraft flies towards steep terrain ●

a. The primary sensor is a downward looking radio altimeter
b. The aural message volume is reduced to one half below 300 feet
c. The alerting boundaries are computed from the barometric height

LEVEL 3 — APPLY AND REASON

330. Rising ground is approached at 2600 ft radio altitude, so why does the basic GPWS stay quiet? ●

a. Its aural alerts are suppressed with the landing gear up
b. Its boundaries are computed from the barometric altitude
c. Its automatic operation begins at a lower height

331. The radio altimeter has failed in flight, so what can still warn the crew about high ground ahead? ●

a. The below glide slope alert of mode 5
b. The enhanced system terrain functions
c. The basic ground proximity modes

332. The crew sees the surrounding high ground pictured on the navigation display at 3000 ft, so which system provides it? ●

a. The weather radar system
b. The enhanced system
c. The basic warning system

333. The limits of one of the alerting envelopes are exceeded, so what does the enhanced system give the flight crew? ●

a. Aural alert messages with visual annunciations and displays
b. An automatic pull-up commanded through the autopilot
c. A master warning that has to be reset by the crew

334. A stable non-precision approach is flown in low visibility with the gear down and the landing flaps out, so why does the early system stay silent?

- **a.** It is inhibited below 500 feet
- **b.** It expects the aeroplane to land
- **c.** It has lost its radio altitude input

335. A system is required to see steep rising ground that the downward looking sensor would find too late, so which protection is needed? ●

- **a.** The modulation of the alerting envelopes
- **b.** The terrain data base look ahead protection
- **c.** The terrain clearance floor protection

336. The aircraft is recognised as landing at one of the troublesome airports held in the database, so what does the system do?

- **a.** It switches off all of its aural warnings
- **b.** It changes over to the terrain display only
- **c.** It adjusts the warning modes for that airport

337. A terrain clearance floor function is found on an aircraft, so which generation of the system is installed? ●

- **a.** The MARK I system
- **b.** The enhanced system
- **c.** The MARK V system

Answers — Ground proximity warning principles

307. a The system warns the crew when the aircraft mistakenly comes too close to the ground. (L1, PDF p. 897)

308. b The warning is an aural alert plus annunciator lights in front of the pilot. (L1, PDF p. 897)

309. a The terrain picture is presented on the navigation display. (L1, PDF p. 899)

310. b The computer, the indications and the inhibit switches are the main parts. (L1, PDF p. 901)

311. b The operative range of the basic GPWS begins at 30 ft radio altitude. (L1, PDF p. 903)

312. a Those two are the additional functions listed for the enhanced system. (L1, PDF p. 903)

313. c The downward looking radio altimeter is the primary sensor for the warning calculation. (L1, PDF p. 905)

314. a The aural alert uses a synthetic voice. (L2, PDF p. 897)

315. c The system is rarely installed in helicopters. (L2, PDF p. 897)

316. c Some aircraft add alert messages on the primary flight displays. (L2, PDF p. 897)

317. c Every aeroplane except the very small ones carries the system. (L2, PDF p. 897)

318. b The enhanced terrain functions do not depend on the radio altitude. (L2, PDF p. 899)

319. a The terrain picture is what the advanced system adds. (L2, PDF p. 899)

320. c Its inputs come from an air data computer and from the navigation systems. (L2, PDF p. 901)

321. a The inhibit switches are the part fitted to some aeroplanes only. (L2, PDF p. 901)

322. a Its stated purpose is to help prevent CFIT accidents. (L2, PDF p. 903)

323. a The aircraft position is compared with an internal database. (L2, PDF p. 903)

324. a The position comes from the FMS or the GPS. (L2, PDF p. 903)

325. b Windshear alerting is mode 7. (L2, PDF p. 903)

326. a The parameters are taken in and alerting algorithms are applied to them. (L2, PDF p. 903)

327. b The two late 1980s products were MARK V for digital and MARK VII for analogue aircraft. (L2, PDF p. 905)

328. a Terrain display and terrain data base look ahead protection are the named improvements. (L2, PDF p. 905)

329. a The warning is calculated from a downward looking sensor, which sees steep terrain late. (L2, PDF p. 905)

330. c The aircraft is still above the radio altitude at which the basic system begins to operate automatically. (L3, PDF p. 897)

331. b The enhanced terrain functions work independently of the radio altitude, while the basic modes need it. (L3, PDF p. 899)

332. b The enhanced system shows the terrain around the aircraft on the navigation display and gives a warning when the aircraft gets too near the terrain, and those functions are independent of the radio altitude. (L3, PDF p. 899)

333. a Exceeding an envelope limit produces aural messages together with visual annunciations and displays. (L3, PDF p. 903)

334. b In the landing configuration the early system expects a landing and issues no warning. (L3, PDF p. 905)

335. b Look ahead protection works from the database instead of the downward looking sensor. (L3, PDF p. 905)

336. c At such an airport the warning modes are adjusted, and only under specific conditions. (L3, PDF p. 905)

337. b The terrain clearance floor belongs to the enhanced system only. (L3, PDF p. 905)

GPWS modes 1 to 5

PDF pages 907–917 · 53 questions

LEVEL 1 — RECALL

338. Which unit is the control unit of the EGPWS ●

a. The display management computer
b. The air data computer
c. The enhanced computer

339. Which condition does mode 1 alert on ●

a. An excessive descent rate
b. An excessive terrain closure rate
c. An altitude loss after take-off

340. Which aural message is generated when the aircraft penetrates the outer alert boundary of mode 1 ●

a. SINK RATE
b. DON'T SINK
c. PULL UP

341. Which condition does mode 2 alert on ●

a. An excessive terrain closure rate
b. An unsafe clearance from the terrain
c. An excessive descent rate

342. For how long after take-off is the mode 2B configuration valid ●

a. For the first 90 seconds
b. For the first 30 seconds
c. For the first 60 seconds

343. Which condition does mode 3 alert on ●

a. An unsafe clearance from terrain
b. An excessive rate of descent
c. An altitude loss after take-off

344. A DON'T SINK message is heard in the flight deck, so which mode has generated it? ●

a. Mode 3
b. Mode 1
c. Mode 4

345. Which condition does mode 4 alert on ●

a. An unsafe clearance from the terrain
b. An excessive rate of descent
c. An altitude loss after the take-off

346. How many submodes does mode 4 have ●

a. Four
b. Two
c. Three

347. Which condition does mode 5 alert on ●

a. A deviation below the glide slope beam
b. An unsafe clearance from the terrain
c. An excessive rate of descent near ground

348. How many levels of mode 5 operation does the computer provide ●

a. Two
b. Three
c. One

LEVEL 2 — UNDERSTAND AND EXPLAIN

349. Which additional protection does mode 6 provide

a. A comparison of the aircraft position with a terrain database
b. A selectable menu of radio altitude callouts during the approach
c. An alert for an excessive rate of closure with the terrain

350. Which part of the system's operation is independent of the other functions ●

a. The warning function
b. The terrain awareness display function
c. The built-in test equipment function

351. The descent continues after the first mode 1 message and the inner alert boundary is penetrated, so what is heard then? ●

a. TERRAIN TERRAIN
b. TOO LOW TERRAIN
c. PULL UP

352. Which three inputs define the two alert boundaries of mode 1 ●

a. Barometric vertical speed, inertial vertical speed and radio altitude
b. Barometric altitude, glide slope deviation and gear position
c. Radio altitude, computed airspeed and the position of the flaps

353. What does the computer produce besides the aural message in both mode 1 cases ●

a. A command to the autopilot to level the aircraft
b. A terrain picture on the navigation display
c. A discrete output to drive the visual annunciations

354. In which flight conditions can the aircraft be when a mode 2 alert is generated ●

a. In the landing configuration only
b. In descent, level flight or climb
c. In descent or level flight only

355. A mode 2 alert boundary is being computed, so which two signals are combined? ●

a. Vertical speed and radio altitude
b. Airspeed and the position of the flaps
c. Barometric altitude and glide slope deviation

356. When is the mode 2A alert generated ●

a. With the flaps set to the landing position on an approach
b. In any configuration other than the one defined for 2B
c. Within the first sixty seconds following the take-off

357. Which aural message comes first when the aircraft penetrates the mode 2A alerting boundaries ●

a. TERRAIN TERRAIN
b. SINK RATE
c. PULL UP

358. The aircraft goes on penetrating the mode 2A boundaries after the first message, so which message is repeated until it leaves the envelope? ●

a. DON'T SINK
b. TERRAIN TERRAIN
c. PULL UP

359. What makes the upper alert boundary of mode 2A vary ●

a. The rate of closure with the terrain
b. The position of the landing flaps
c. The speed of the aircraft

360. Within how many dots of the localizer or glide slope centreline does the mode 2B configuration apply during an ILS approach ●

a. Within plus or minus 3 dots
b. Within plus or minus 2 dots
c. Within plus or minus 1 dot

361. Which messages are suppressed when both gear and flaps are in the landing position in mode 2B ●

a. The TERRAIN messages
b. The PULL UP messages
c. The TOO LOW GEAR messages

362. In which two flight phases does mode 3 provide its alerts ●

a. During the final approach and the landing roll
b. During the cruise and during a precision approach
c. Immediately after take-off and during a missed approach

363. Until when do the mode 3 visual annunciators remain active ●

a. Until the landing gear has been retracted
b. Until the warning envelope has been left
c. Until a positive rate of climb is established

364. Which configuration must exist after take-off before the computer initiates a mode 3 alert

a. The gear down and the flaps at landing
b. The gear up and the flaps not at landing
c. The gear up and the flaps set to landing

365. By reference to what does mode 4 detect an insufficient terrain clearance ●

a. The glide slope and the localizer
b. The phase of flight and the speed
c. The vertical speed and the bank angle

366. When is mode 4A active ●

a. During take-off with the gear and the flaps still retracted
b. During an ILS approach with the flaps in landing position
c. During cruise and approach with the landing gear not at landing

367. At which speed does a mode 4A penetration with the landing gear up give the TOO LOW GEAR message ●

a. At 250 knots and above
b. At 190 knots or below
c. At 159 knots or below

368. The airspeed builds up during the cruise, so what happens to the mode 4A upper boundary? ●

a. It decreases linearly
b. It stays constant
c. It increases linearly

369. In which configuration does mode 4B give the TOO LOW FLAPS message ●

a. With the gear down and the flaps not at landing
b. With the gear and the flaps both still fully retracted
c. With the gear up below one hundred and fifty nine knots

370. On what is mode 4C based ●

a. A fixed boundary of five hundred feet during the approach
b. A deviation of the flight path below the glide slope beam
c. A minimum terrain clearance floor that increases during take-off

371. Mode 5 alerts are not given on every kind of approach, so on which one are they given? ●

a. A front course ILS approach
b. A non-precision radio approach
c. A back course ILS approach

372. How loud is the soft alert GLIDE SLOPE message ●

a. The same as the other alert message volume levels
b. About one half of the other alert volume levels
c. About twice the other alert message volume levels

373. A hard alert activation occurs on an approach, so below which radio altitude is the aircraft? ●

a. 500 feet
b. 245 feet
c. 300 feet

374. How far below the glide slope must the aircraft be for the hard alert activation ●

a. More than 3 dots
b. More than 1.3 dots
c. More than 2 dots

LEVEL 3 — APPLY AND REASON

375. The terrain awareness display function has been lost, so what happens to the ground proximity warning functions? ●

a. They fall back to the excessive descent rate mode
b. They are lost together with the display function
c. They go on working while the necessary inputs are there

376. An installation carries only the modes that are basic system requirements, so which basic GPWS alerting mode is missing? ●

a. Mode 6
b. Mode 5
c. Mode 4

377. A warning lamp stays dark although the aural message is heard, so which computer function is suspect?

a. The terrain awareness display
b. The input signal processing
c. The alert output processing

378. A replacement enhanced computer is being fitted, so what has to be set so that it works with the right modes?

a. The inhibit switches, before the first flight
b. The maintenance pages, after the installation
c. The program pins, before it is installed

379. The crew reduces the descent rate as soon as the first mode 1 message is heard, so which boundary has not been penetrated? ●

a. The upper alert boundary
b. The inner alert boundary
c. The outer alert boundary

380. Both the landing gear and the flaps are in the landing position when the mode 2B envelope is penetrated, so which message is repeated? ●

a. PULL UP
b. TERRAIN TERRAIN
c. TERRAIN

381. The mode 2B envelope is penetrated with the landing gear down but the flaps still up, so which message is heard first? ●

a. TERRAIN TERRAIN
b. TOO LOW TERRAIN
c. PULL UP

382. The flaps are at the landing position and the aircraft is within two dots of the localizer on an ILS approach, so which submode boundary applies? ●

a. Mode 2A
b. Mode 2B
c. Mode 4B

383. A go-around is flown with the gear up and the flaps away from landing when a larger altitude loss than permitted is detected, so which alert is initiated? ●

a. The mode 4 alert
b. The mode 3 alert
c. The mode 1 alert

384. A computer has to decide how much height may be lost before a mode 3 alert is given, so what does it do?

a. It reads the loss from the terrain database
b. It calculates the amount of permitted loss
c. It compares the loss with a fixed boundary

385. Which mode uses altitude above sea level information in its computation ●

a. Mode 3
b. Mode 1
c. Mode 2

386. The mode 4A boundary is penetrated at 250 knots with the landing gear still retracted, so which message does the computer generate? ●

a. TOO LOW GEAR
b. TOO LOW TERRAIN
c. TOO LOW FLAPS

387. Just after take-off with gear and flaps up the radio altitude falls below the long term filter value, so which message is generated? ●

a. TOO LOW GEAR
b. TOO LOW TERRAIN
c. DON'T SINK

388. An approach is flown at 140 knots with the gear down and the flaps still up, so which submode watches the terrain clearance? ●

a. Mode 4C
b. Mode 4B
c. Mode 4A

389. The aircraft drifts 1.5 dots below the beam centre at 800 feet radio altitude, so which activation occurs? ●

a. The soft alert activation
b. No alert activation at all
c. The hard alert activation

390. A glide slope deviation message is heard noticeably quieter than the other warnings, so what does that tell the crew? ●

a. It is the hard alert activation
b. It is a failure of the loudspeaker
c. It is the soft alert activation

Answers — GPWS modes 1 to 5

338. c The EGPWC is the control unit of the system. (L1, PDF p. 907)

339. a Mode 1 is the excessive descent rate mode. (L1, PDF p. 909)

340. a The outer boundary gives the SINK RATE message. (L1, PDF p. 909)

341. a Mode 2 covers an excessive rate of closure with the terrain. (L1, PDF p. 911)

342. c Mode 2B applies for the first 60 seconds after take-off. (L1, PDF p. 911)

343. c Mode 3 is the altitude loss after take-off mode. (L1, PDF p. 913)

344. a Mode 3 gives the DON'T SINK message. (L1, PDF p. 913)

345. a Mode 4 covers unsafe terrain clearance. (L1, PDF p. 915)

346. c Mode 4 has three submodes. (L1, PDF p. 915)

347. a Mode 5 covers a deviation below the glide slope. (L1, PDF p. 917)

348. a There are two levels of mode 5 operation. (L1, PDF p. 917)

349. b Mode 6 offers selectable radio altitude callouts on the landing approach. (L2, PDF p. 907)

350. a The warning function is the one stated to be independent of the others. (L2, PDF p. 907)

351. c The inner boundary gives the PULL UP message. (L2, PDF p. 909)

352. a Those three inputs define the two mode 1 boundaries. (L2, PDF p. 909)

353. c A discrete output drives the visual annunciations in both cases. (L2, PDF p. 909)

354. b Mode 2 alerts are possible in descent, in level flight and in the climb. (L2, PDF p. 911)

355. a Vertical speed and radio altitude are the two signals used. (L2, PDF p. 911)

356. b Mode 2A covers every configuration that is not a mode 2B configuration. (L2, PDF p. 911)

357. a The first mode 2A message is TERRAIN TERRAIN. (L2, PDF p. 911)

358. c PULL UP is repeated until the aircraft exits the warning envelope. (L2, PDF p. 911)

359. c The upper boundary of this mode varies with aircraft speed. (L2, PDF p. 911)

360. b The tolerance is plus or minus 2 dots. (L2, PDF p. 911)

361. b In that configuration the PULL UP messages are suppressed. (L2, PDF p. 911)

362. c The two phases are immediately after take-off and during a missed approach. (L2, PDF p. 913)

363. c They stay active until a positive rate of climb exists. (L2, PDF p. 913)

364. b The alert needs the gear up and the flaps away from the landing position. (L2, PDF p. 913)

365. b Detection is made with reference to the phase of flight and the speed. (L2, PDF p. 915)

366. c Mode 4A covers cruise and approach with the gear not in the landing configuration. (L2, PDF p. 915)

367. b The speed for that message is 190 knots or below. (L2, PDF p. 915)

368. c The boundary increases linearly with airspeed. (L2, PDF p. 915)

369. a That message belongs to the gear down, flaps not at landing case. (L2, PDF p. 915)

370. c Mode 4C uses a minimum terrain clearance floor that rises during take-off. (L2, PDF p. 915)

371. a Mode 5 works on a front course ILS approach. (L2, PDF p. 917)

372. b The soft alert is at approximately one half, minus 6dB, of the other levels. (L2, PDF p. 917)

373. c The hard alert belongs to radio altitudes below 300 feet. (L2, PDF p. 917)

374. c The hard alert needs more than 2 dots below the glide slope. (L2, PDF p. 917)

375. c The warning functions keep working as long as their inputs remain available. (L3, PDF p. 907)

376. a There are six basic modes, of which only modes 1 to 5 are basic system requirements, so mode 6 is the one an installation limited to the requirements leaves out. (L3, PDF p. 907)

377. c The warning lamp drivers belong to the alert output processing. (L3, PDF p. 907)

378. c The modes are selected by program pins before the computer is installed. (L3, PDF p. 907)

379. b The first message comes from the outer boundary, so the inner one has still not been crossed. (L3, PDF p. 909)

380. c With gear and flaps at landing the repeated message is TERRAIN. (L3, PDF p. 911)

381. a With either gear or flaps not at landing the initial message is TERRAIN TERRAIN. (L3, PDF p. 911)

382. b That configuration is one of those listed for mode 2B. (L3, PDF p. 911)

383. b That is exactly the condition for the mode 3 alert. (L3, PDF p. 913)

384. b The computer itself calculates the permitted loss before alerting. (L3, PDF p. 913)

385. a Mode 3 is the mode whose computation is stated to take altitude above sea level information, while mode 1 works from vertical speed and radio altitude and mode 2 from vertical speed and radio altitude. (L3, PDF p. 913)

386. b With the gear still up the boundary is penetrated at 190 knots or below as TOO LOW GEAR, while the boundary that rises with airspeed to 1000 feet at 250 knots gives TOO LOW TERRAIN. (L3, PDF p. 915)

387. b That decrease below the filter value gives TOO LOW TERRAIN. (L3, PDF p. 915)

388. b Mode 4B is the submode that applies once the landing gear is lowered. (L3, PDF p. 915)

389. a The aircraft is beyond 1.3 dots below the beam centre, so the soft alert comes; the hard alert needs below 300 feet radio altitude and more than 2 dots. (L3, PDF p. 917)

390. c Only the soft alert is given at about half the normal volume. (L3, PDF p. 917)